Highlights

When is a piston good enough? Limits of radiation simplifications for shell optimization and numerical benchmarks

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- A piston (sinc) model is judged by whether its designs reach a freely chosen target.
- At $kR = 18.3$ the piston model errs under 0.18 dB yet misses an example -20 dB target.
- At $kR \leq 3$ its designs reach -27 to -42 dB, though overstated by up to 31 dB.
- For an example target of -20 dB, the rule accepted exactly the 13 that reach it.
- Robust design under 1% thickness errors does not close the gap to exact designs.

When is a piston good enough? Limits of radiation simplifications for shell optimization and numerical benchmarks

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Abstract

Shell optimization often estimates radiated sound power with the sinc matrix, exact only for a flat baffled surface. A piston model is enough when its design reaches the true power level that the project demands, relative to a reference design, even if the model misstates the power itself. A design is accepted only when its predicted level plus the largest error measured for the model lies at or below the target; the rule holds for any target, and we illustrate it with two arbitrary choices, −20 and −40 dB. We test a steel spherical cap with 960 thickness variables against a two-sided reference, exact for the prescribed-velocity, zero-thickness open cap. At 100 Hz ($kR = 18.3$) the piston model predicts its three end designs within 0.18 dB, yet its uniform-start design reaches only −17.1 dB, against −26.6 dB for the exact model. In six cases on caps of 30° to 120° half-angle, scaled through the sound speed to $kR = 1$ to 7, piston-driven designs reach −26.8 to −41.9 dB, although at $kR \leq 3$ their reductions are overstated by up to 31 dB; exact-driven designs reach −53 to about −80 dB. An error budget measured on other cases of its kR class accepted, for the −20 dB example, all 13 of the 14 tested end designs that reach it and rejected the one that misses it. Under independent 1% thickness errors, robust piston designs remain 9.1 and 26.8 dB above robust exact designs at $kR = 18.3$ and 3.

Keywords: Radiated sound power, Radiation matrix, Spherical cap,

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Structural-acoustic optimization, Robust design, Error budget, Numerical benchmark

1. Introduction

A shell can vibrate with a small amplitude and still radiate a considerable amount of acoustic power. When the task is to make it quieter by changing its thickness, the designer needs the radiated power of every candidate design, and its gradient, many hundreds of times. The power is a quadratic form in the normal velocity of the surface; its matrix, the radiation matrix, holds the acoustic interaction between every pair of surface points. Radiation matrices are established tools for sound-power prediction and active control [1–4], and for objective functions in structural-acoustic design [5, 6]; acoustic topology optimization has also been carried out with finite-element models of the fluid domain [7].

The cheapest radiation matrix is exact for a flat surface in an infinite rigid baffle, with kernel $\sin(kr)/(kr)$; k is the acoustic wavenumber and r the distance between two surface points. We call this the sinc kernel, or piston model: "piston" names the baffled-plane reference, not a uniform-velocity assumption for the shell. Its smooth, real kernel avoids a boundary-element solve, and it has been applied unchanged to curved shells without a baffle, including in the thesis on which this paper builds [8]. That changes the physical problem: a free open shell has fluid on both faces and an edge around which fluid can move. For flat plates, the difference between baffled and unbaffled radiation has been studied in detail [9, 10]; approximate power models have also been compared with full acoustic solutions for engineering structures [11–13]. Such comparisons judge a model by accuracy.

A designer asks a different question. Choosing an objective function decides which design an optimizer reaches, as O. M. Silva and co-workers showed for the dynamic compliance and for the active and reactive input power of one-material structures under steady-state forced vibration [14, 15]. In the same spirit, we evaluate whether the piston model is *enough*: a radiation model is good enough when its design attains the level of radiated power that the project demands, relative to a reference design, even when the model misstates the power itself. Modern solvers can reduce the radiated power of a design down to -80 dB relative to the reference, but they rely on sensitive physical and mathematical properties to attain those levels, which are very difficult to manufacture and to observe in reality.

Two difficulties stand in the way of a plain answer. The first is that the prediction error e of the piston model (the true level minus the level predicted by the piston model) is not a constant: it depends on frequency, opening angle and velocity pattern. An optimizer seeks out the designs where the model errs in its favor, leaving a selection gap s (the number of decibels by which the piston-driven design radiates more than the best design known under the exact model). We therefore budget the error instead of ignoring it, and accept a design only when its predicted level, plus the largest error measured for the model, still lies at or below the target level. A subtler issue concerns the designs that reach the lowest levels. They owe their quietness to cancellation between parts of the surface, and a small disturbance, such as the added mass of the surrounding air, can multiply their radiated power by up to 29, although our exact solver resolves the deepest of these designs only to about 2 dB in vacuo and 30% with fluid loading. A target attained only by a fragile design is not attained in practice, which calls for an objective that averages over manufacturing errors [16, 17], as Valentini, O. M. Silva and Cardoso did for the harmonic displacement of structures under an uncertain excitation frequency [18].

Thus, we ask: *when does a piston radiation model produce a design that reaches a prescribed level of the true radiated power, and does that design survive the fluid and the tolerances of manufacture?*

We study a family of steel spherical caps with a 10 m radius and half-angles of 30°, 90° and 120°, driven at the apex and supported at the rim (Fig. 1 illustrates the 30° design point). The acoustic reference is exact for the prescribed-velocity, zero-thickness, two-sided problem on each open cap. We solve the open-cap dual-series equations with a Galerkin method, following work on scattering by open spherical shells [19–21]. Reducing the solution to structural coefficients avoids a boundary-element solve for each velocity field and lets us use the same adjoint optimizer.

The design point is the 30° cap in air at 100 Hz, $kR = 18.3$. Six regime cases span the three half-angles and vary an effective sound speed, while the structure keeps its 100 Hz drive and the specific impedance keeps its value in air; they scale one problem rather than sweep the operating frequency. The error budget and its pass rates are therefore measured on this cap family. Another geometry, frequency, load or support needs its own calibration on representative optimizer paths and its own held-out test, by the same procedure.

The main finding is that accuracy at the end design does not decide

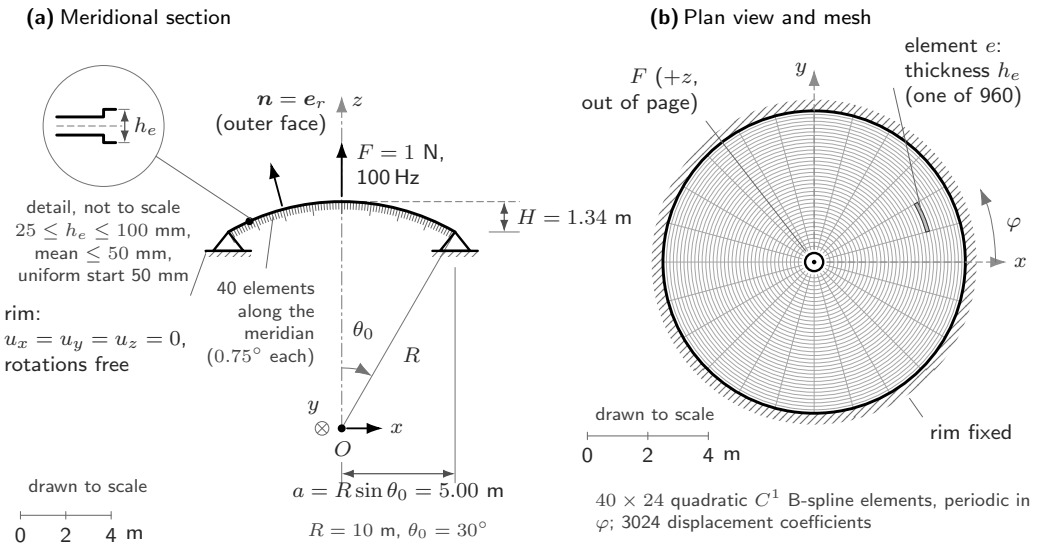


Figure 1: The steel spherical cap of this paper. (a) Meridional section: radius of curvature $R = 10$ m, half-angle $\theta_0 = 30°$, rim radius $a = R \sin\theta_0$ and rise H; apex force $F = 1$ N at 100 Hz along the axis; pinned rim ($u_x = u_y = u_z = 0$, rotations free); the unit normal $\boldsymbol{n}$ points to the outer face; inset: thickness of one element, not to scale. (b) Plan view with the 40×24 quadratic C^1 B-spline elements (periodic around the axis), each with its own thickness h_e. Both panels are drawn to scale.

whether the target is reached. At the design point the sinc matrix rates its end designs within 0.18 dB, yet its descent from the uniform shell stops short of -20 dB; at low kR it overstates the reduction by tens of decibels, yet its designs still reach -20 dB.

In each of the seven cases, each model starts from uniform thickness and from the end design produced by the other model. At the design point, it also starts from the earlier sinc design G. The acceptance rule works for any target T; we apply it at $T = 20$ and 40 dB, two values we chose arbitrarily to illustrate it, and test it on piston-driven designs in vacuo and with fluid loading. We also test an expected-power objective under independent thickness errors.

All two-sided results share one acoustic idealization: radiation from the shell mid-surface, no acoustic thickness, equal normal velocity on both faces, unbounded fluid, and no cavity. Shell thickness still enters the structural model through mass and stiffness; it is the design variable and the quantity perturbed in the robust objective.

Section 2 defines the cap and model; Section 3 defines the acceptance rule; Section 4 tests it. The remaining sections report robust and fluid-loaded designs, reproduction limits and conclusions; the appendices give full settings, verification and secondary tests.

2. The problem and the setup

Figure 2 summarizes the computation; the comparisons below change only the radiation operator. Table 1 collects the model settings, and Table A.1 in Appendix A lists those of the operators, objectives and optimizer.

2.1. The object

The object is a thin steel shell shaped as a spherical cap, drawn to scale in Fig. 1: panel (a) shows what holds and drives it, panel (b) what the optimizer may change, and Table 1 lists its settings. With $R = 10$ m and $\theta_0 = 30°$, its rim radius is $a = R\sin\theta_0 = 5$ m and its rise is $H = R(1 - \cos\theta_0) = 1.34$ m. A harmonic force of 1 N drives the apex at 100 Hz, along the axis and away from the center of the sphere; the rim is pinned and acoustically transparent, so the fluid moves freely around it. The shell radiates into air, and at 100 Hz $kR = 18.32$ and $ka \approx 9.2$. The optimizer may change the thickness h_e of each of the 960 elements, between 25 and 100 mm, with the mean thickness held at or below 50 mm; the uniform design, $h_e = 50$ mm everywhere, is the baseline and the cold start. Each patch in the thickness maps of Fig. 5 represents one of these variables; the 3,024 displacement coefficients and the acoustic quadrature points are separate from the design variables.

2.2. Shell model and structural response

The shell is modeled with Kirchhoff–Love isogeometric elements [22, 23]: quadratic C^1 B-splines on a mesh of 40 elements along the meridian and 24 around the axis (Table 1). The element arrays are computed once, so that the stiffness and mass matrices are reassembled for any thickness field as $\mathbf{K}(h) = \sum_e (h_e^3 \mathbf{K}_e^{\mathrm{b}} + h_e \mathbf{K}_e^{\mathrm{m}})$ and $\mathbf{M}(h) = \sum_e h_e \mathbf{M}_e$, with bending and membrane parts $\mathbf{K}_e^{\mathrm{b}}$ and $\mathbf{K}_e^{\mathrm{m}}$. For a thickness field h, the discrete equilibrium at angular frequency ω is:

$$\mathbf{K}_D(h,\omega)\,\mathbf{q} = \mathbf{f}, \qquad \mathbf{K}_D = \mathbf{K}(h) - \omega^2\mathbf{M}(h) + i\omega\mathbf{C}(h), \tag{1}$$

where $\mathbf{C} = \beta\mathbf{K}$ is the damping matrix, with $\beta = 0.1/\omega$ and no mass-proportional term, $\mathbf{f}$ is the load vector and $\mathbf{q}$ holds the displacement coefficients, with the time factor $e^{i\omega t}$. Because $\mathbf{K}_D$ is complex symmetric, one

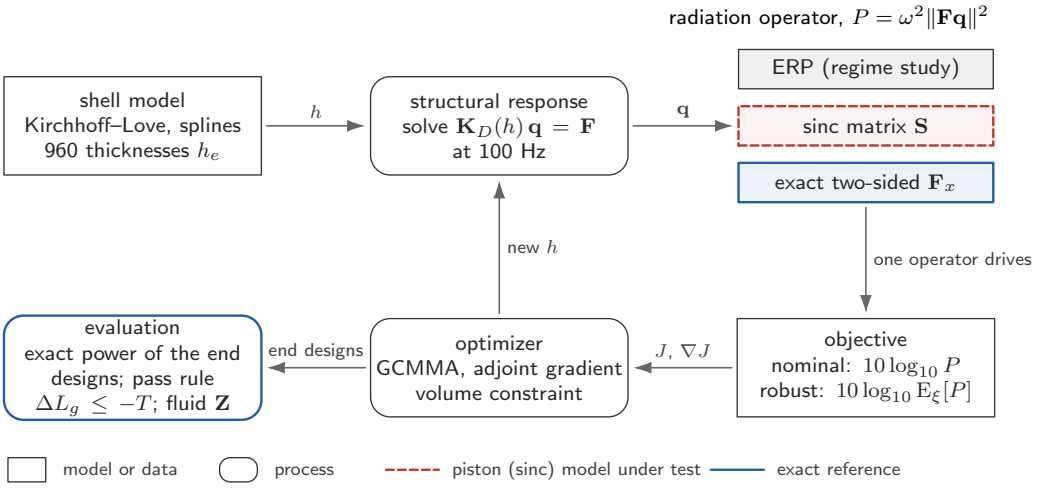


Figure 2: The computational chain behind every result. One radiation operator drives the optimization loop, which runs from the structural response through the operator and the objective to the optimizer and back; the end designs then leave the loop and are evaluated with the exact two-sided operator, in vacuo and with the fluid-loading operator $\mathbf{Z}$, against the pass rule $\Delta L_g \le -T$. Grey: phase-blind ERP (regime study only); red dashed: the piston (sinc) model under test; blue: the exact reference. Sharp boxes are models or data, rounded boxes are processes. Schematic.

sparse LU factorization per design serves the state solve and the adjoint solve. The acoustic operators need the normal velocity $v_n = i\omega\,\mathbf{n}\cdot\mathbf{u}$ of the mid-surface, which at the acoustic quadrature points is a linear map, $\mathbf{v} = i\omega\mathbf{Bq}$, with 4×4 points per element (15 360 in all) for the sinc operator and 8×8 (61 440) for the exact operator and ERP. Fluid loading is left out of Eq. (1) except in Section 6; the specific acoustic impedance of air, $\rho_0 c_0 = 415$ Pa s/m, is about 0.17% of the mass impedance of the uniform shell, $\omega m'' = 2.5\times10^5$ Pa s/m with $m'' = 393$ kg/m^2, an estimate that ignores stiffness and radiation reactance.

2.3. Radiation operators

Three operators turn the velocity into power, each in the form $P_M = \omega^2\|\mathbf{F}_M\mathbf{q}\|^2$, where $\mathbf{F}_M$ is the acoustic factor for operator M; it is distinct from the structural load vector $\mathbf{f}$ in Eq. (1). Algorithm 1 lists how each factor is built.

Equivalent radiated power (ERP). The phase-blind measure $P_{\mathrm{ERP}} = \frac{1}{2}\rho_0 c_0 \int_\Gamma |v_n|^2\,dS$ treats every point of the surface Γ as a piston in its own

Table 1: Settings of the model, in the order of the chain of Fig. 2. The radiation operators, the fluid-loading operator, the objectives, the optimizer, the stopping rule and the thread counts are listed in Table A.1.

Item	Setting
Geometry	spherical cap, $R = 10$ m, $\theta_0 = 30°$ ($90°$ and $120°$ in the regime study), mid-surface only
Shell model	Kirchhoff–Love, quadratic C^1 B-splines; 40×24 elements (meridian $\times$ circumference), periodic around the axis; 3024 displacement coefficients, 2952 free; element arrays with 4×4 Gauss points per element
Material, damping	steel, $E = 200$ GPa, $\nu = 0.3$, 7850 kg/m^3; $\mathbf{C} = \beta\mathbf{K}$, $\beta = 0.1/\omega$ (loss factor 0.1), no mass-proportional damping
Load, supports	1 N force at the apex along the axis, in $+z$, away from the center of the sphere; 100 Hz; shared equally by the 24 control points of the apex row; rim: the rim row of control points fixed in all three directions (displacements fixed, rotations free); rim acoustically transparent
Fluid	air, $c_0 = 343$ m/s, $\rho_0 = 1.21$ kg/m^3, $kR = 18.32$; regime study: $c' = \omega R/kR$ with $\rho' c' = \rho_0 c_0$
Design variables	960 element thicknesses, $25 \le h_e \le 100$ mm; uniform design and cold start $h_e = 50$ mm
Constraint	mean element thickness at most 50 mm, $g = \overline{h}/(0.05\ \text{m}) - 1 \le 0$ (elements not weighted by area)

baffle, with no interaction between points. This local approximation cannot credit cancellation, and it serves here as the cheapest objective.

The piston (sinc) model. For a flat patch in a rigid baffle, the Rayleigh integral gives the pressure, and the time-averaged power becomes a symmetric double integral:

$$P_{\text{sinc}} = \alpha \int_\Gamma \int_\Gamma \overline{v_n(\mathbf{x})}\, j_0(k|\mathbf{x}-\mathbf{y}|)\, v_n(\mathbf{y})\, dS_y\, dS_x, \qquad \alpha = \frac{\rho_0 \omega^2}{4\pi c_0}, \tag{2}$$

where $j_0(s) = \sin(s)/s$. Applied with source points on a curved shell, Eq. (2) defines what we call the sinc model; this form equals twice the power of one free-space layer of monopoles (Appendix B). We use the spline-consistent matrix $\mathbf{S}$, which takes the field and the normal from the spline basis at every quadrature point. The code never forms $\mathbf{S}$ itself: instead, j_0 is written as

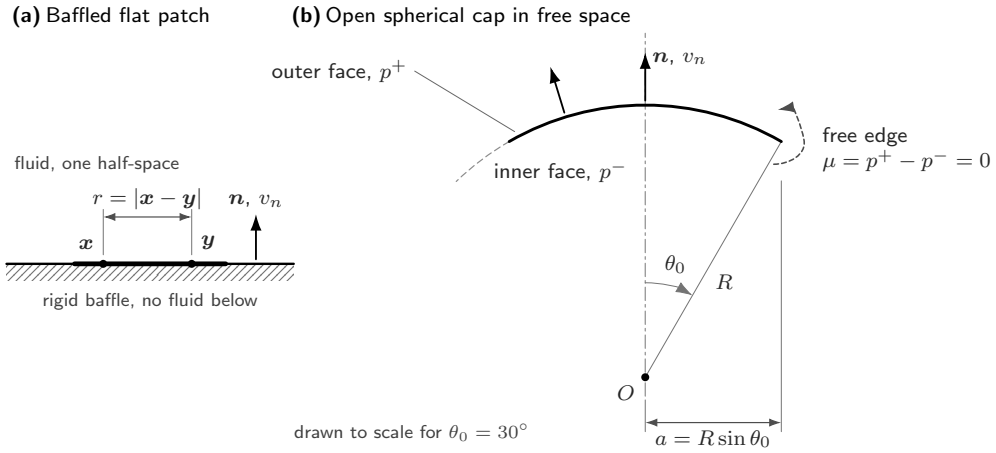


Figure 3: The two acoustic problems compared in this paper. (a) Flat patch in an infinite rigid baffle, radiating into one half-space; for this configuration the sinc kernel $j_0(k|\boldsymbol{x}-\boldsymbol{y}|)$ gives the exact power. (b) Open spherical cap in free space, the problem solved here. Both faces touch the fluid, the unit normal $\boldsymbol{n}$ points to the outer face, and the pressure difference $\mu = p^+ - p^-$ vanishes at the free edge. Schematic; (b) is drawn to scale for $R = 10$ m and $\theta_0 = 30^\circ$.

an average of plane waves over all directions, so that $\mathbf{S} = \mathbf{F}_{\mathrm{S}}^{\mathrm{H}}\mathbf{F}_{\mathrm{S}}$ up to the quadrature over directions, with one row of $\mathbf{F}_{\mathrm{S}}$ per direction (Appendix B).

The exact two-sided operator. In this zero-thickness, no-cavity reference, the shell mid-surface has the same prescribed normal velocity on both faces. The exterior acoustic solve determines the pressure jump $\mu = p^+ - p^-$, which vanishes at the free edge. The surface integral of this jump's work against v_n gives the radiated power. The hypersingular integral equation below is the numerical form of that solve:

$$(\mathcal{H}\mu)(\mathbf{x}) = i\omega\rho_0\, v_n(\mathbf{x}) \text{ on } \Gamma, \qquad \mu = 0 \text{ on the free edge,} \tag{3}$$

where $\mathcal{H}$ is the normal derivative of the double-layer potential, written with the time factor $e^{-i\omega t}$ of the acoustic solver. The radiated power is:

$$P_{2s} = \tfrac{1}{2}\,\mathrm{Re}\int_\Gamma \mu\,\overline{v_n}\,dS. \tag{4}$$

On a sphere, the double-layer operator is diagonal in spherical harmonics: with $x = kR$, the coefficients of the radial velocity and of the pressure jump obey $\rho_0 c_0 v_{\ell m} = \lambda_\ell \mu_{\ell m}$, $\lambda_\ell = x^2 j_\ell'(x) h_\ell'(x)$, where $h_\ell = h_\ell^{(1)}$ is the spherical

Hankel function of the first kind. A cap requires μ to vanish on the rest of the sphere while v_n is prescribed on the cap. These dual series equations [19, 20] are solved by a Galerkin method whose basis functions vanish like the square root of the distance to the edge, 48 per azimuthal order m with $|m| \le 20$; the truncation degree $\ell_{\max}$ of the operator sum is removed by Richardson extrapolation from $\ell_{\max} = 4000$ and 8000. Because the power is a Hermitian positive semidefinite form in the velocity, the solution can be reduced once to the structural coefficients, $P_{2s} = \omega^2 \|\mathbf{F}_x \mathbf{q}\|^2$; the factor took 54 s to build and reproduces the direct solution on six frozen designs to within 0.07%. Its Hermitian blocks were computed in two algebraically equal forms, one for the design-point runs and a round-off-safe one for the regime study and the fluid loading, which give the same power to seven digits at the design point. A boundary-element solution agrees with the exact one to 0.1–0.6% on identical velocity traces, and the flat-disc limit to between 3×10^{-5} and 5.4×10^{-4} (Appendix C). The reference does not, however, resolve designs that cancel to below about 2×10^{-3} of the power of the designs combined, each normalized to unit exact power; where deeper designs appear, their resolution is stated with them.

2.4. Objectives

The nominal objective is the radiated power of the design, $J(h) = P_M(h) = \omega^2 \|\mathbf{F}_M \mathbf{q}(h)\|^2$ in watts. The optimizer receives it as a scaled level,

$$f_0(h) = 100\,\frac{A + 10\log_{10} J(h)}{p_0}, \qquad A = 100, \qquad p_0 = A + 10\log_{10} J(h^{(0)}), \tag{5}$$

so that $f_0 = 100$ at the start design $h^{(0)}$; this is the scaling of the thesis code [8], kept unchanged. It increases with J only while $p_0 > 0$, that is, while the start radiates more than 10^{-10} W; for a quieter start, minimizing f_0 would raise the power. The design-point runs use the operators as they are, whose powers at the uniform design are 3.8×10^{-7} W (sinc) and 5.9×10^{-7} W (exact). In the regime study, where the powers of the uniform design range down to 3×10^{-11} W and warm starts can be far quieter, each operator is multiplied by a constant so that the start design, uniform or warm, has $J = J_{\mathrm{ref}} = 5.9464 \times 10^{-7}$ W, the exact power of the uniform design at the design point; then $p_0 = 37.7$. The re-optimization with fluid loading and the robust runs are rescaled in the same way. A constant factor changes neither the minimizers nor the levels, although the path of the optimizer can depend

Algorithm 1 Building the radiation operators of one case $(\theta_0,\, kR)$.

Input: spline shell of Section 2.2; wavenumber k; fluid impedance $\rho_0 c_0$
Output: factors $\mathbf{F}_{\mathrm{E}}$ (ERP), $\mathbf{F}_{\mathrm{S}}$ (sinc), $\mathbf{F}_x$ (exact); fluid-loading matrix $\mathbf{Z}$

sample the surface: points $\mathbf{x}_q$, weights w_q, normals $\mathbf{n}_q$; build $\mathbf{B}$ with $\mathbf{v} = i\omega\mathbf{B}\mathbf{q}$
ERP: $\mathbf{F}_{\mathrm{E}} \leftarrow \mathrm{diag}\big(\sqrt{\rho_0 c_0\, w_q/2}\big)\,\mathbf{B}$ ▷ 8×8 points per element
sinc: directions $\mathbf{s}_d$ from 32 Gauss–Legendre nodes in $\cos\vartheta$ times 64 azimuths, weights a_d summing to one; row d of $\mathbf{F}_{\mathrm{S}}$: $\sqrt{\alpha a_d}\,\sum_q e^{-ik\mathbf{s}_d\cdot\mathbf{x}_q} w_q \mathbf{B}_q$ ▷ 4×4 points
for each azimuthal order $|m| \le 20$ **do**
    project the normal displacement on the 48 edge functions: Galerkin right-hand sides $\mathbf{G}_m\mathbf{q}$ ▷ 8×8 points
    whitening $\mathbf{T}_m$: drop Gram eigenvalues below 10^{-12} of the largest (35 to 48 functions kept)
    Galerkin matrix in the whitened basis, $\mathbf{A}_m(\ell_{\max})$, for $\ell_{\max} = 4000$ and 8000
    $\mathbf{H}_m(\ell_{\max}) \leftarrow \frac{1}{4}\rho_0 c_0\,(\mathbf{A}_m^{-1} + \mathbf{A}_m^{-\mathrm{H}})$ ▷ design point
        or $\mathbf{H}_m(\ell_{\max}) \leftarrow \frac{1}{2}\rho_0 c_0\,\mathbf{A}_m^{-1}(\mathrm{Re}\,\mathbf{A}_m)\mathbf{A}_m^{-\mathrm{H}}$ ▷ regime study, fluid loading
    Richardson: $\mathbf{H}_m \leftarrow 2\mathbf{H}_m(8000) - \mathbf{H}_m(4000)$; eigendecompose, set negative eigenvalues to zero
    append the rows $\mathbf{\Lambda}_m^{1/2}\mathbf{V}_m^{\mathrm{H}}\mathbf{T}_m\mathbf{G}_m$ to $\mathbf{F}_x$
    fluid loading: $\mathbf{Z} \leftarrow \mathbf{Z} + \mathbf{Q}_m^{\mathrm{H}}\mathbf{Z}_m\mathbf{Q}_m$, $\mathbf{Q}_m = \mathbf{T}_m\mathbf{G}_m$, with $\mathbf{Z}_m$ the Richardson step of $2\mathbf{H}_m(\ell_{\max}) + \frac{1}{2}\rho_0 c_0(\mathbf{A}_m^{-1} - \mathbf{A}_m^{-\mathrm{H}})$ (second form of $\mathbf{H}_m$)
conjugate $\mathbf{Z}$ to the structural time factor $e^{i\omega t}$; for the re-optimization, symmetrize it on the free coefficients

on it (Appendix G). The robust objective replaces J by the expected power under random relative thickness errors,

$$J_{\mathrm{rob}}(h) = \mathrm{E}_\xi\big[P\big(h \odot (\mathbf{1} + \sigma\boldsymbol{\xi})\big)\big] \approx \frac{1}{N}\sum_{i=1}^{N} P\big(h \odot (\mathbf{1} + \sigma\boldsymbol{\xi}_i)\big), \tag{6}$$

where $\odot$ is the element-wise product, the ξ_e are independent standard normal variables, one per element, and $\sigma = 0.01$ is the standard deviation of the relative error. The 25–100 mm bounds apply to nominal h only; the Gaussian perturbations are not clipped or projected back into that interval. The right-hand side is the sample-average approximation with $N = 8$ samples in the main runs ($N \in \{8, 16, 32\}$ in the sample-size check of Section 5), drawn once from a fixed normal bank (seed 4242) and kept for the whole run, so that the objective is a smooth deterministic function of h. The sample average

is taken over radiated powers in watts rather than decibel levels, and enters Eq. (5) in place of J. Its gradient follows from the adjoint of each sample by the chain rule, $\partial J_{\mathrm{rob}}/\partial h_e = N^{-1}\sum_i (1+\sigma\xi_{ie})\,\partial P_i/\partial h_e$. Section 5 gives the sample-size study and results.

2.5. Optimizer

The optimizer is our MATLAB port of Svanberg's GCMMA [24], with at most 600 outer iterations (several thousand in the continued runs of Section 4.4), three settings changed from the defaults of the implementation and no move limit (Table A.1). Algorithm 2 writes out the loop, which with $N = 1$ and $\sigma = 0$ is the nominal optimization. A run stops when the windowed change of f_0, the largest thickness change and the constraint violation stay below their thresholds for three consecutive iterations, or at the iteration limit, and it returns the iterate with the lowest power, which for a run still descending is not the last one (run X0 returned a design with a mean thickness of 47.2 mm). The adjoint gradients agree with central differences to 3.6×10^{-7} (exact) and 3.9×10^{-7} (sinc) at the design point, and to 1.9×10^{-7}–1.6×10^{-6} for all three operators in the regime study (Appendix A).

2.6. Evaluation and runs

Whichever operator drove the run, its designs are evaluated with the sinc and exact operators, in the regime study also with ERP, and the exact power is the verdict (Algorithm 3). At the design point every iterate of the six runs is evaluated, 3121 designs with the six frozen fields; in each case of the regime study, every fifth iterate and the returned design of each run, the uniform design and 40 random smooth thickness fields (Appendix A). Checks of resolution and an independent spectral evaluation of the sinc power accompany the evaluation, and the evaluation with fluid loading of Section 6 covers the uniform and returned designs only. Cross-evaluation separates two quantities that a single run mixes: what the driving model predicted, and what the design actually radiates.

Algorithm 2 Optimization loop, nominal ($N = 1$, $\sigma = 0$) or robust.

Input: driving acoustic factor $\mathbf{F}_M$; start $h^{(0)}$; σ; samples $\boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_N$ (fixed seed); limit $k_{\max}$ (600; 150 with fluid loading; up to 5000 in Section 4.4)

Output: returned design h^* (lowest J) and the history of every iterate

scaled runs: $\mathbf{F}_M \leftarrow \mathbf{F}_M \sqrt{J_{\text{ref}}/J(h^{(0)})}$

$p_0 \leftarrow 100 + 10 \log_{10} J(h^{(0)})$ ▷ must be positive

for $k = 1, \ldots, k_{\max}$ **do**

  for $i = 1, \ldots, N$ **do**

    $h_i \leftarrow h^{(k-1)} \odot (\mathbf{1} + \sigma \boldsymbol{\xi}_i)$; solve $\mathbf{K}_D(h_i)\mathbf{q}_i = \mathbf{f}$ ▷ robust: GMRES, preconditioned by the nominal LU

    $P_i \leftarrow \omega^2 \|\mathbf{F}_M \mathbf{q}_i\|^2$; adjoint solve with the same matrix; ∇P_i with respect to h_i

  $J \leftarrow \frac{1}{N} \sum_i P_i$; $\nabla J \leftarrow \frac{1}{N} \sum_i (\mathbf{1} + \sigma \boldsymbol{\xi}_i) \odot \nabla P_i$

  if $J < J^*$ **then** $J^* \leftarrow J$, $h^* \leftarrow h^{(k-1)}$

  $f_0 \leftarrow 100\,(100 + 10 \log_{10} J)/p_0$; $g \leftarrow \overline{h}/(0.05\,\text{m}) - 1 \leq 0$

  GCMMA step (at most 20 inner iterations, each with the true f_0 and g) $\rightarrow h^{(k)}$; store $h^{(k)}$

  if $k \geq 21$ and the stopping rule of Table A.1 has held three times in a row **then stop**

Algorithm 3 Evaluation of the designs of one case.

Input: designs d (iterates, returned designs, random fields); baseline d_0 (uniform); factors $\mathbf{F}_{\text{E}}, \mathbf{F}_{\text{S}}, \mathbf{F}_x$; $\mathbf{Z}$

for each design d and the baseline **do**

  solve Eq. (1); $P_M \leftarrow \omega^2 \|\mathbf{F}_M \mathbf{q}\|^2$ for $M \in \{\text{S}, \text{x}\}$, and E in the regime study

  resolution: repeat P_x with $\ell_{\max} = 8000$ alone and, in the regime study, with 32 Galerkin functions; check: sinc power from its spherical-harmonic series

  fluid loading (uniform and returned designs): solve $(\mathbf{K}_D + i\omega\alpha_f \mathbf{Z})\mathbf{q} = \mathbf{f}$, $\alpha_f = 1$ (scaled medium) or c'/c_0 (air density); P_x again

levels $\Delta L_M(d)$, error $e(d)$ and gap $s(d)$ of Section 3; compare ΔL_x with $-T$

At the design point, each operator was run from the uniform thickness, from an earlier sinc-optimized design G (Appendix D), and from the end design of the other operator's run from uniform; the runs are named by operator (S, sinc; X, exact) and start (0, uniform; G; S or X, the other run's end), so that S0 and X0 are the two cold starts. The regime study of Section 4.2 repeats a similar plan in seven cases with ERP added, and Section 6 re-optimizes one case with fluid loading in the loop.

3. What "enough" means

3.1. Target and level

For a model M, the level of a design d is its power relative to the baseline d_0, here the uniform thickness, in decibels and negative when the design is quieter:

$$\Delta L_M(d) = 10 \log_{10} \frac{P_M(d)}{P_M(d_0)}. \tag{7}$$

We write $\Delta L_{\text{pred}} = \Delta L_{\text{sinc}}$ for the level that the piston model predicts and $\Delta L_x = \Delta L_{2s}$ for the true one, given by the exact operator. A project may select any reduction $T > 0$ relative to the reference design d_0; its target level is $-T$, and a design reaches that target when $\Delta L_x(d) \leq -T$. The rule below is generic in T. In this study we choose $T = 20$ dB and $T = 40$ dB as illustrative decision levels; neither is a universal or physically privileged threshold. Levels near -60 to -80 dB, which the exact optimizer reaches in some cases below, rely on sensitive physical and mathematical properties, which are very difficult to manufacture and to observe in reality; we report them as secondary results.

3.2. Error budget and set-conditional level

The piston model errs, and the error of a level is what matters to the designer. It is defined here as the true minus the predicted level,

$$e(d) = \Delta L_x(d) - \Delta L_{\text{pred}}(d) = 10 \log_{10} \frac{\rho(d_0)}{\rho(d)}, \qquad \rho(d) = \frac{P_{\text{sinc}}(d)}{P_{2s}(d)}, \tag{8}$$

so that $e > 0$ means that the design is louder than the piston model predicts, that is, the model *overstates* its quietness, and $e < 0$ that it understates it. The second form follows from the definitions: a large absolute error does no harm to a level if the ratio ρ of the design and that of the baseline are the same.

Let $e_{\max}$ be the largest error measured over a set $\mathcal{D}$ of designs evaluated with the reference, $e_{\max} = \max_{d \in \mathcal{D}} e(d)$. For every design of that set, Eq. (8) gives $\Delta L_x(d) \leq \Delta L_{\text{pred}}(d) + e_{\max}$. We therefore call

$$\Delta L_g = \Delta L_{\text{pred}} + e_{\max} \tag{9}$$

the set-conditional level (called the guaranteed level below). This bound applies only to designs in $\mathcal{D}$, not to arbitrary designs or optimizer paths.

The decision test is generic in T: we accept a piston-produced design only when $\Delta L_g \leq -T$, so its predicted level plus the largest measured kernel error remains at or below the target. The test can be used for any target T, but empirical coverage is conditional on the calibration set, path family and prediction range; the held-out checks here evaluate only $T = 20$ dB and $T = 40$ dB. For the robust objective, ΔL_{pred} is the level of mean power predicted by the piston model, $10 \log_{10}\{\mathrm{E}_\xi[P_{\mathrm{sinc}}(d \odot (1 + \sigma\xi))]/P_{\mathrm{sinc}}(d_0)\}$; it is not the mean of the sample dB levels, $\mathrm{E}_\xi[\Delta L_{\mathrm{pred}}(d \odot (1 + \sigma\xi))]$. By Jensen's inequality, $10 \log_{10} \mathrm{E}[P] \geq \mathrm{E}[10 \log_{10} P]$, so the level of mean power is higher (less quiet) and gives the cautious comparison with a quieter-than-target requirement. For a nominal design without thickness errors, it reduces to $\Delta L_{\mathrm{pred}}(d)$. A model can meet the target even when its prediction is inaccurate, provided the target leaves room for the error.

The S0 run at the design point ($kR = 18.3$) illustrates the rule. Its predicted level is $\Delta L_{\mathrm{pred}} = -17.217$ dB; holding out the design-point case, the fit to the 90° and 120° cases at $kR = 7$ gives $e_{\max} = 0.900$ dB, so $\Delta L_g = -17.217 + 0.900 = -16.317$ dB (about -16.3 dB). This case-specific budget differs from the pooled high-kR line in Table I.1. Since -16.3 dB is above the -20 dB target, the rule rejects S0, whose true level, -17.1 dB, also misses that target. The low-kR numerical example is given in Section 4.3, after its envelope has been defined.

This is a conditional bound: it applies only to designs used to measure $e_{\max}$, and an optimizer may leave that set. No design-independent bound exists. Over all designs evaluated for the 30° cap at $kR = 1$, the ratio ρ spans 1.8×10^{-3} to 3.5×10^{5} (5th to 95th percentile), so e itself spans more than 80 dB there. The budget is calibrated on designs similar to those the optimizer visits: piston-driven iterates from other cases of the same class, fixed before outcomes are examined (Section 4.3). Random thickness fields, which the cheap model did not produce, do not qualify.

3.3. Three kinds of error, kept short

Three measures of error recur below, one for each use a designer makes of a model. The absolute error is the ratio ρ, which matters only when the absolute level is needed. On frozen velocity fields it is far from constant: for six cap fields it lies 2 to 3 orders of magnitude above one when the cap is acoustically small, falls to minima of 0.15 to 0.25 near $kR = 3$–4, and oscillates about 0.5 above $kR \approx 10$ (Appendix D). The level error e of Eq. (8) depends only on how ρ changes from the baseline to the design, and it enters

the budget. The selection gap judges a design by its exact power against the best design known under the exact model, d^*:

$$s(d) = 10 \log_{10} \frac{P_{2s}(d)}{P_{2s}(d^*)} = \Delta L_x(d) - \Delta L_x(d^*). \tag{10}$$

It measures how far the design lies above the best one known, the decibels left on the table. Because d^* comes from local runs that did not all converge, the true optimum is at least as quiet, and s is a lower bound. This gap serves as a secondary result: a design with a large s can still reach the target.

3.4. Robustness

A target is reached in practice only if it survives the differences between the model and the built shell. The deepest designs of this paper owe their quietness to cancellation, and a disturbance of a few parts in 10^4 of their kinetic energy, the added mass of the fluid, undoes a large part of it (Section 6). We therefore apply the rule also to the expected power under thickness errors, Eq. (6), and design for it directly in Section 5.

4. Does the piston reach the target?

4.1. The design point

At 100 Hz, the sinc-driven runs S0 (uniform start) and SG (design-G start) both finish at −17.1 dB, above the −20 dB target. The exact-driven uniform-start run X0 reaches −26.6 dB; no run reaches the other target, −40 dB (Table 2). The piston model was accurate all the same: it predicted the levels of its own designs S0, SG and SX to within 0.18 dB, because their ratios, 0.61 to 0.66, stay close to that of the uniform design, 0.636. It even predicted correctly that S0 would fall short of the −20 dB target. At the design point the piston model is accurate but, from the uniform start, falls short of this illustrative target.

Only one run, SG, met the stopping rule, after 109 iterations. Of the five runs that stopped at the 600-iteration limit, three changed by at most 0.006 dB over their last 100 iterations and run XS by 0.05 dB; run X0 was still descending by 0.15 dB per 100 iterations, so the local optimum along its path is quieter than the design it returned. Continued from that design (Section 4.4), X0 met the stopping rule at iteration 4355, at −26.87 dB, only 0.27 dB lower; the gaps of Table 2 would grow by that amount.

Table 2: Cap optimization at 100 Hz under the sinc and exact operators. ΔL_{pred}, ΔL_x: level relative to the uniform design predicted by the sinc model and true (exact), Eq. (7), in dB, negative when quieter; ρ: sinc-to-exact power ratio (0.636 for the uniform design); e: true minus predicted level, Eq. (8), positive when the design is louder than predicted; s: level above the best exact design found (X0), Eq. (10), a lower bound because X0 had not converged. Run names give the operator (S, sinc; X, exact) and the start. The last row is the earlier sinc-optimized design G.

Run	Operator	Start	Iterations	ΔL_{pred}	ΔL_x	ρ	e	s
S0	sinc	uniform	600	−17.22	−17.09	0.618	+0.13	9.50
SG	sinc	design G	109	−17.21	−17.03	0.609	+0.18	9.57
SX	sinc	end of X0	600	−25.69	−25.87	0.662	−0.18	0.73
X0	exact	uniform	600	−24.78	−26.59	0.964	−1.81	0 (ref.)
XG	exact	design G	600	−17.01	−17.43	0.701	−0.42	9.16
XS	exact	end of S0	600	−17.17	−17.84	0.742	−0.67	8.75
G	—	(earlier)	—	−17.21	−17.04	0.612	+0.17	9.55

From uniform, S0 and X0 descend together for about 100 iterations to roughly −18 dB; X0 then crosses the −20 dB target while S0 levels off above it (Fig. 4(a)). Panel (b) compares predicted and true levels at the end; Table 2 summarizes the four other starts.

Started at X0, the sinc optimizer stayed within 1 dB of it in both models and reached a predicted level of −25.7 dB. The shortfall therefore comes from the path taken from the uniform start rather than how the two end designs are rated: the piston model retains the deeper minimum, but its descent from the uniform start did not reach it. Because this comparison uses one cold start per operator with a local method, it does not imply that the piston model misses deeper designs from every starting point.

On the shared scale in Fig. 5, S0 places a solid 100 mm disc at the apex out to about 1.05 m of arc; X0 stays at the 25 mm lower bound to about 0.5 m and places its 100 mm material in a ring from about 0.5 to 1.2 m. Near the rim the order reverses: X0 carries thicker bands than S0 (ring means of 59 to 85 mm between 4.5 and 4.9 m of arc, against 48 to 51 mm for S0 on the same rings).

The exact-driven design X0 radiates inefficiently. Its exact radiation efficiency, $\sigma_{\mathrm{rad}} = P_{2s}/(\frac{1}{2}\rho_0 c_0 \int_\Gamma |v_n|^2\, dS)$, is 0.0105, against 0.73 for the uniform design and 0.34 for S0, although its mean-square normal velocity is 3.7 times that of S0. Figure 6 shows why: out-of-phase surface contributions nearly

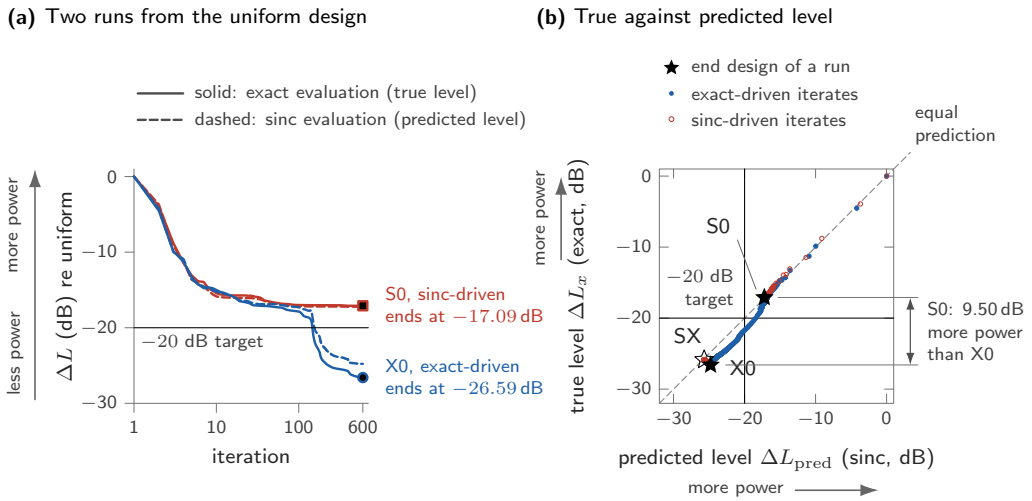


Figure 4: Optimization paths under the sinc and exact operators at 100 Hz; levels relative to the uniform design, higher means more power. (a) Level against iteration for the two runs from the uniform design, S0 (sinc-driven) and X0 (exact-driven); every iterate is evaluated with both operators: solid, exact (the true level); dashed, sinc (the predicted level). The horizontal line marks the −20 dB example target. (b) True against predicted level for every iterate of the six runs, with the end designs of S0, SX and X0 starred; the −20 dB example target lines are drawn on both axes. The arrow gives the difference in true level between the end designs of S0 and X0, from one start with a local method. Five of the six runs stopped at the 600-iteration limit, while run SG met the stopping rule at iteration 109.

cancel, so X0 radiates little despite its larger vibration amplitude.

The two-sided operator credits such cancellation more than the piston model does: the ratio of X0 is 0.96 against 0.64 for the uniform design, so the piston model places X0 1.81 dB above its true level ($e = -1.81$ dB), an error on the safe side for the rule of Section 3.2. Near the end designs the two operators disagree on direction as well: from X0 the sinc optimizer lowered its own objective by 0.91 dB while the exact power rose by 0.73 dB, and the rank correlation between the two operators, 0.92 to 0.99 along the descent, falls to near zero or below among the designs close to the end designs (Appendix E).

4.2. A regime study

The design point sits where ρ changes little from design to design. We repeated the comparison in seven cases, fixed before any result was computed:

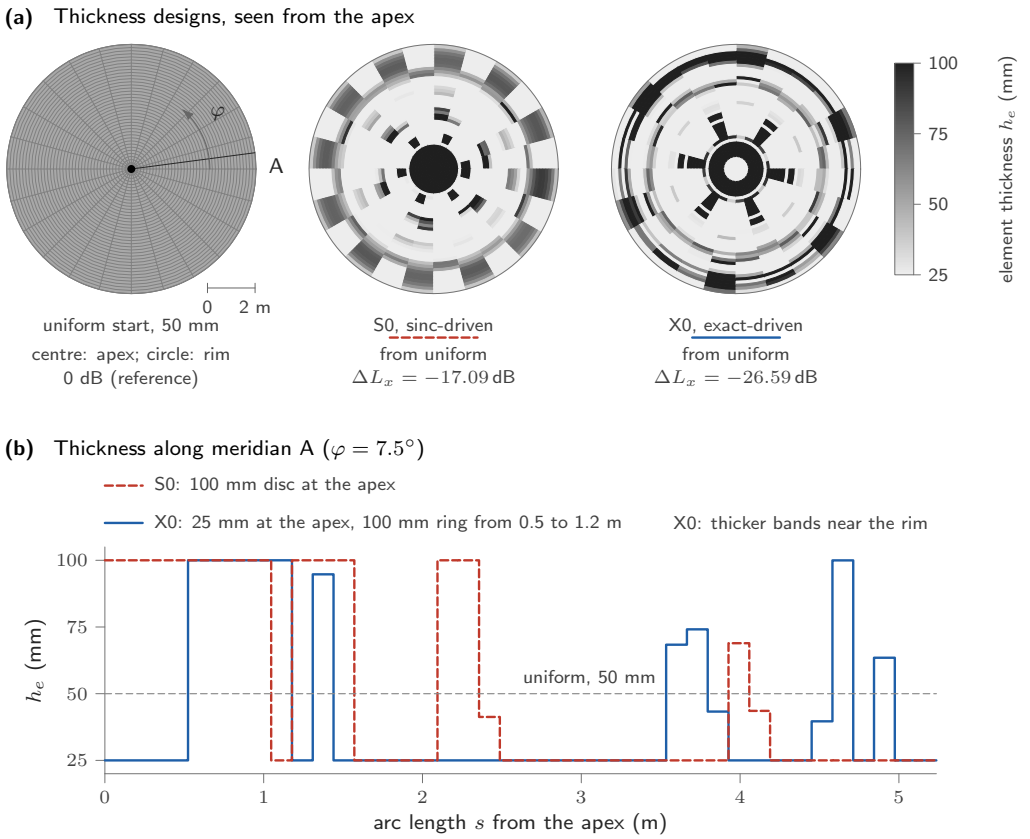


Figure 5: Optimized thickness designs at the design point, both started from the uniform 50 mm design. (a) Thickness of each element, drawn flat without smoothing, on a map of the cap centered on the apex, whose radius is the meridian arc length from the apex; one gray scale from 25 to 100 mm for all three maps. From left to right: the uniform start, with the mesh and meridian A; the end designs of runs S0 (sinc-driven) and X0 (exact-driven), both iteration-limited; under each map, its true level ΔL_x relative to the uniform design. (b) Thickness along meridian A ($\varphi = 7.5°$).

the 30° cap at $kR = 1$ and 3 and at the design point, and caps of 90° and 120° at $kR = 1.5$ and 7. Only the half-angle and the acoustic wavenumber change, and the structure is always driven at 100 Hz, where each cap has modes below the drive frequency (the first natural frequency of the uniform design is 79.3, 44.9 and 23.4 Hz for the three openings). A resonant steel cap cannot be brought to kR of order one by scaling R and the frequency together, since at a fixed nondimensional frequency kR is fixed by the material and the fluid. We set kR through an effective sound speed instead, $c' = \omega R/kR$,

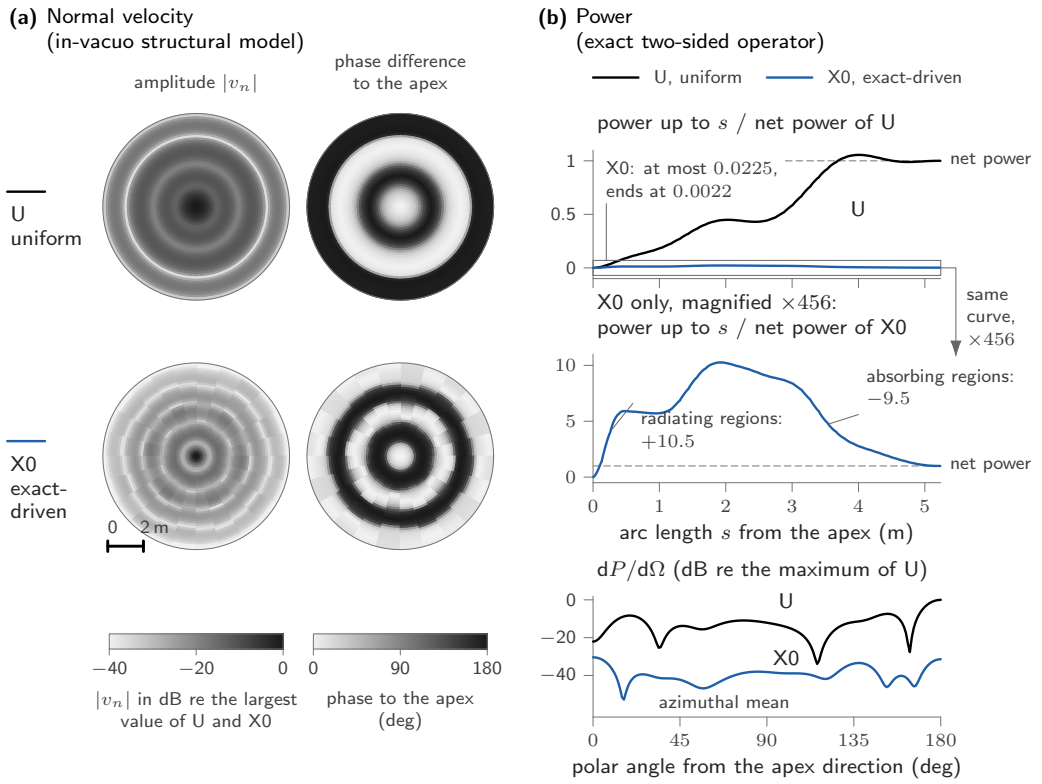


Figure 6: How the exact-driven design X0 reaches its low level, set against the uniform design U, at 100 Hz. (a) Normal velocity on the same map as Fig. 5: element-mean amplitude in dB relative to the largest value of both designs, and phase difference to the apex ring (0, in phase; 180°, antiphase). (b) Top: power radiated by the cap from the apex up to arc length s, both designs divided by the net power of U. Compare with the uniform design: X0 radiates a small fraction of the power of U (0.22%), and its curve never exceeds 0.0225 at any s. Middle: the same X0 curve divided by the net power of X0, which magnifies it 456 times. It shows where the cancellation of X0 happens; regions where the curve rises radiate, regions where it falls absorb. Bottom: azimuthal mean of the far-field power per solid angle against the polar angle from the apex direction, both designs in dB relative to the maximum of U. In-vacuo structural response; exact two-sided radiation with $\ell_{\max} = 8000$.

and kept the specific impedance of air, $\rho' c' = \rho_0 c_0$; this amounts to changing the ratio of the longitudinal wave speed of the shell to c' from 15.4 (steel in air) to 0.84, 2.5 and 5.9 for $kR = 1$, 3 and 7. The six cases away from the design point are therefore scaled problems. In each case all three objectives were run from the uniform design, the sinc and ERP objectives were run a

Table 3: Regime study on seven cases of half-angle θ_0 and kR, all driven at 100 Hz; levels relative to the uniform design and gaps in dB. Best exact ΔL_x: true level of the quieter exact-driven design; sinc ΔL_x: true level of the sinc-driven design from the uniform start; sinc e: its true minus predicted level, Eq. (8), positive when louder than predicted; sinc s: its level above the best exact design found in the case, Eq. (10), a lower bound. All sinc and exact runs stopped at the 600-iteration limit (continued in Section 4.4). The last row is the design point of Table 2. The spread between the two exact-driven runs and the gaps of the ERP designs are given in Appendix G.

θ_0	kR	best exact ΔL_x	sinc ΔL_x	sinc e	sinc s
30°	1	−78.0	−41.9	30.8	36.0
90°	1.5	−60.1	−28.4	19.0	31.7
120°	1.5	−53.1	−26.8	23.5	26.2
30°	3	−84.5[a]	−34.3	21.6	50.3
90°	7	−44.9	−39.2	1.5	5.7
120°	7	−42.9	−39.4	0.8	3.6
30°	18.3	−26.6	−17.1	0.1	9.5

[a] Below what the reference resolves, about −80 dB in this case (Section 4.4).

second time from the end design of the exact run, and the exact objective was run a second time from the end design of the sinc run, giving two runs per objective in each case (42 runs in all, Table 3); 34 of them stopped at the 600-iteration limit, and the eight that met the stopping rule were all ERP runs. Section 4.4 continues the sinc and exact runs of five cases beyond that limit.

Across the six scaled cases, the sinc-driven uniform-start designs reach true levels from −26.8 to −41.9 dB; the 30°, $kR = 1$ case also reaches the −40 dB target (Table 3). At $kR \leq 3$ it did so although the piston model placed its designs 19.0 to 30.8 dB lower than their true level; at $kR = 7$ it was within 1.5 dB. The predictions alone would have misled a designer at the −40 dB target: the piston model predicted reductions beyond that level for the sinc-driven design in all six scaled cases (predicted levels −40.1 to −72.7 dB), whereas the true level reached it in one. The error budget is needed because the prediction alone is optimistic. ERP never reached the −20 dB target outside the two 30° cells at low kR (−24.4 and −22.7 dB); on the 120° cap at $kR = 1.5$, its design is 0.06 dB louder than the uniform one (Eq. 9).

In Fig. 7(a), each sinc design's gap to the best exact design is compared with the exact optimizer's 10 dB spread between starts. Panel (b) shows

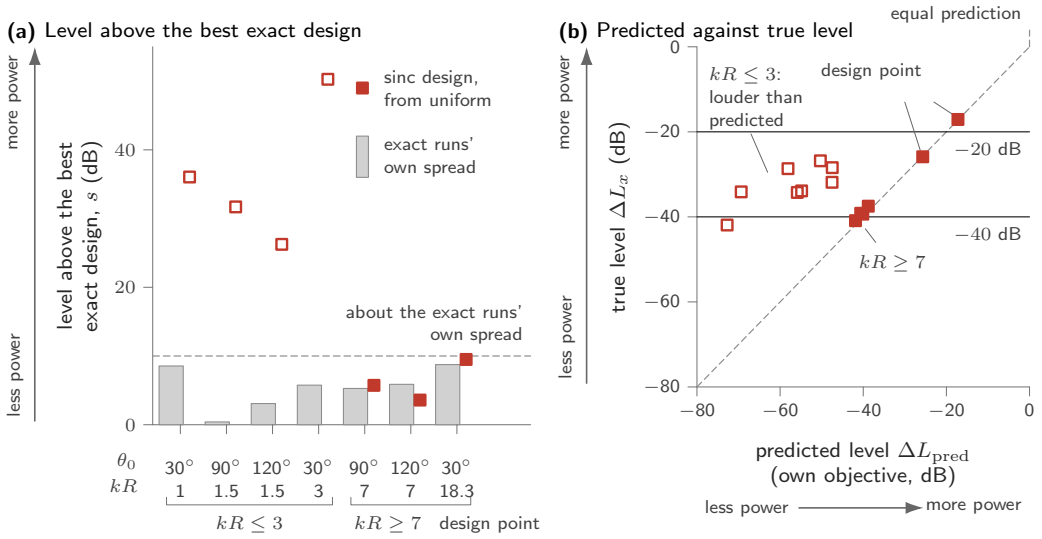


Figure 7: Sinc-driven designs of the regime study, in the seven cases ordered by kR. (a) Level above the best exact design, s, of the sinc design from the uniform start (squares; open for $kR \leq 3$, filled for $kR \geq 7$); higher means more power. The gray bars give the spread between the two exact-driven runs, and the dashed line at 10 dB is about that spread; s is a lower bound. (b) Predicted against true level for the 14 sinc designs, from the uniform start and from the end of the exact run, both relative to the uniform design; a point above the line of equal prediction is louder than predicted. Horizontal lines show the example targets −20 and −40 dB. All sinc and exact runs stopped at the 600-iteration limit. Zero-thickness two-sided model; kR set through the sound speed, structure driven at 100 Hz.

acceptance at the −20 dB target and the difference between predicted and true levels. The ERP and exact-driven runs are shown in the same way in Appendix G.

At $kR \leq 3$, the best exact designs reach −53 dB to at least about −80 dB. The deepest designs on the 30° cap at $kR = 3$ lie below the reference's resolution; the sinc-driven designs lie 26.2 to 50.3 dB above their exact counterparts, with the upper gap dependent on that unresolved depth. Their exact efficiencies are 2×10^{-11} to 10^{-8}, reached by cancellation that the piston model rates differently (Appendix G). The exact optimizer's spread between starts is about 10 dB, so the post hoc threshold of 10 dB separates model error from optimizer variability. By this measure the sinc model fails at $kR \leq 3$ for all three openings; no failure appears at $kR \geq 7$, where its gaps are 5.7 and 3.6 dB. By target attainment, the result is less severe: the missed

depths lie at the reference's resolution limit and do not survive fluid loading intact (Section 6).

4.3. The set-conditional level

The rule of Section 3.2 uses two inputs: the predicted level and the designs that supply the error budget. The first follows from the objective of Section 2.4: ΔL_{pred} is the level, in decibels, of the expected power $J = \mathrm{E}[P]$ predicted by the sinc model; every design below is nominal. The budget comes from designs of the kind being certified, the sinc-driven iterates of the other cases of the same class, and it may grow as the predicted level falls through an upper envelope $e_{\max}(\Delta L_{\mathrm{pred}}) = a - b\,\Delta L_{\mathrm{pred}}$, the lowest such line above all of those iterates. The classes are set in advance by kR: for descriptive pooled summaries, the low-kR fit has $a = 5.1$ dB and $b = 0.435$, and the high-kR fit has $a = 0.05$ dB and $b = 0.037$; decisions use only the separate case-held-out refits below. The design point belongs to the high-kR class, where e lies between -2.0 and $+1.1$ dB over all 3160 evaluated designs. The pooled coefficients and population totals in Table I.1 summarize class data; they do not generate the case-specific budgets below.

For each test case, its cell is held out and the envelope is refitted using sinc-driven iterates from the other cases in the same class. The low-kR class contains the $30°/kR = 1$, $90°/kR = 1.5$, $120°/kR = 1.5$, and $30°/kR = 3$ cases; the high-kR class contains $90°/kR = 7$, $120°/kR = 7$, and the $30°/kR = 18.3$ design point. Thus the S0 test excludes all design-point data and is calibrated on only the two other high-kR cases; its stored case-specific envelope gives $e_{\max} = 0.900$ dB.

Which designs supply the budget decides the outcome (Appendix I). Random smooth thickness fields stay within about 5 dB of the uniform design, where the error is small, and give an $e_{\max}$ of at most 2.63 dB even in the cases where the sinc model places its own designs 19 to 35 dB lower than they are; with that budget the guarantee held for only 2 of the 14 piston-driven designs. The exact-driven iterates have a negative largest error at $kR \leq 3$, so a budget taken from them would put the guaranteed level below the prediction instead of above it. Nor does a constant budget serve at low kR: along the sinc paths the error grows by 0.42 to 0.55 dB per decibel by which the predicted level falls, so that about half of every extra predicted decibel is not delivered. With the envelope, certifying the -20 dB target at $kR \leq 3$ needs a predicted level of -44.4 dB or lower; certifying the -40 dB target needs -79.8 dB, lower than any sinc run predicted. At $kR \geq 7$ and

at the design point the error is small and nearly flat, so that a constant of about 1 dB would serve as well; there, the −20 dB target is certified from a predicted level of −20.8 dB.

For example, at $kR = 3$ the uniform-start sinc run SU predicts −55.9 dB. The low-kR envelope just defined, with $a = 5.09$ dB and $b = 0.435$, gives $e_{\max} = 29.4$ dB and $\Delta L_g = -26.5$ dB; the design passes the −20 dB target, and its true level is −34.3 dB.

Figure 8 applies the rule to the 14 piston-driven end designs of the seven cases, the runs from the uniform design (U) and from the end design of the exact run (X). The bars hang down from the level of the uniform design, so a longer bar is a quieter design. In each pair, the hatched bar is the set-conditional ceiling ΔL_g and the solid bar is the true level ΔL_x: a design passes when its hatched bar ends below one of the two example target lines (−20 or −40 dB), and the bound holds when the hatched bar does not extend below the solid one ($\Delta L_g \geq \Delta L_x$). The thin stem below each hatched bar, rising from the predicted circle ΔL_{pred}, is the error budget $e_{\max}$ added to it.

For these 14 tested in-vacuo designs, the case-held-out envelopes lay above every true level by 0.1–10.1 dB; there was no false accept at either target (−20 or −40 dB) in this set. At $T = 20$ dB, the per-case rule accepted 13 of 14, exactly those whose true levels reached that level. The count includes three evaluations beyond their corresponding fit ranges: both endpoints from uniform and exact starts at 30°, $kR = 1$, and the exact-start endpoint at 120°, $kR = 7$ (Fig. 8); these decisions rely on extrapolation. The 14 endpoints come from correlated paths within this cap family; the six regimes vary opening angle and effective sound speed at the same 100 Hz structural drive, not independent frequency or application cases. This finite within-family check illustrates the protocol but does not validate transfer to a new problem.

Among the plotted U/X runs, S0 at the design point was the only rejection at the −20 dB target ($\Delta L_g = -16.3$ dB; true level −17.1 dB). SG, a separate run from the frozen design G, was also rejected (−16.3 dB guaranteed; −17.0 dB true) but is not one of the 14 plotted SU/SX designs. The plotted warm-started run SX passed (−24.6 dB guaranteed; −25.9 dB true).

At the −40 dB target, the envelope accepted only the 120° cap at $kR = 7$ from the exact start (−40.4 dB guaranteed; −40.9 dB true); evaluating that case extends slightly beyond the fitted predictions. It rejected the 30° cap at $kR = 1$ from the uniform start, although its true level met the corresponding −40 dB boundary. No design at the 100 Hz design point reached this target (Table 2). Under this same-class envelope, the −20 dB target at low kR

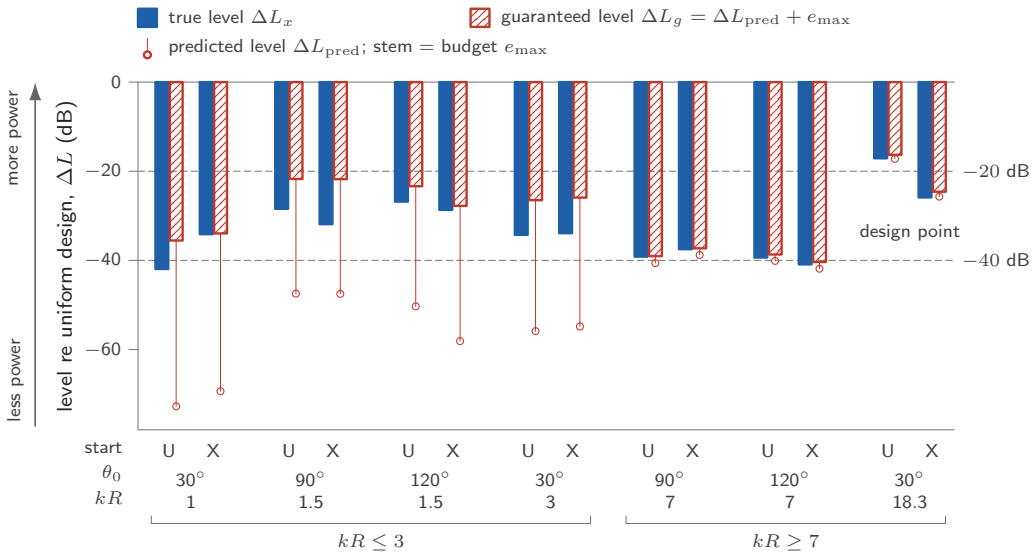


Figure 8: Set-conditional error budget against true level for the 14 piston-driven end designs of the seven cases, started from the uniform design (U) and from the end design of the exact run (X). Levels relative to the uniform design; the bars hang down from 0 dB, and lower means less power. Solid bars: true level ΔL_x (exact, in vacuo). Hatched bars: set-conditional level $\Delta L_g = \Delta L_{\mathrm{pred}} + e_{\max}(\Delta L_{\mathrm{pred}})$, with the envelope fitted on the sinc-driven iterates of the other cases of the same class. This empirical bound is limited to the indicated designs and class; it is not a universal guarantee. Circles: predicted level ΔL_{pred}; the stem from a circle up to its hatched bar is the budget $e_{\max}$. Horizontal lines show the example targets −20 and −40 dB. For any target T, a design passes when its hatched bar ends below $-T$; the bound holds when the hatched bar is not longer than the solid one. For both designs of the 30° cap at $kR = 1$ and for the 120° cap at $kR = 7$ started from X, the envelope is evaluated beyond the range of predictions it was fitted on. At the design point, U and X are runs S0 and SX; the other six cases are scaled, with kR set through the sound speed.

requires a prediction below about −44 dB; at $kR \geq 7$, the margin is about 1 dB.

Each case has two sinc runs, one warm-started at the exact design; neither the envelope nor the design-point mean therefore rests on independent repeats (the mean is discussed in Appendix I). The class envelopes use three or four cases, one load, one drive frequency and one mesh; all cases are scaled except the design point. No case lies between $kR = 3$ and 7, where a new case's class would be ambiguous.

The envelope bounds the observed paths, not every possible path. In the

second stage of strategy S1, a sinc run for the 30° cap at $kR = 3$ started from an ERP design and reached a predicted −58.3 dB with only 6.9 dB error, far below the envelope. Other paths could lie above it. At the design point, an early S0 iterate exceeds the envelope by 0.01 dB; designs B, C and D in Appendix D, optimized for other objectives, exceed it by up to 0.63 dB.

The rule tests target attainment, not optimality. At $kR \leq 3$, it accepts sinc designs meeting the −20 dB target even when the best exact designs beat them by 24 to 50 dB; the upper gap reaches a depth unresolved by the reference (Tables 3 and K.1). The reference truth here is in vacuo. With fluid loading at air density, the bound holds for only 7 of 14 designs: the fluid raises sinc-design true levels by up to 6.1 dB at $kR \leq 3$ and 3.6 dB at $kR = 7$, exceeding the calibrated margins.

4.4. Longer runs and staged strategies

Most sinc and exact runs ended at the 600-iteration limit, so their levels and gaps belong to truncated descents. We continued three runs per case from iteration 600 to 3000 on the 90° and 120° caps at $kR = 1.5$ and the 30° cap at $kR = 3$: exact runs from uniform and sinc starts, and the sinc run from uniform. The exact run X0 of the design point went on until it met the stopping rule. For the 30° cap at $kR = 1$, the stored XS and SU 3000-iteration budget checkpoints remain unaudited and are excluded; only the frozen 1800-iteration results are used for these starts. GCMMA was restarted at iteration 600 with fresh asymptotes, since its state had not been stored, and the counts below include the first 600 iterations. Table 4 reports these outcomes.

At the design point, X0 gained 0.27 dB after iteration 600 over almost 3800 further iterations (Fig. 9(a)). The exact runs from the uniform design met the stopping rule in all five cases, after 944 to 4355 iterations, and in three of them most of the descent had happened by iteration 600: besides the design point, they went about 0.7 dB lower on the 90° cap and on the 30° cap at $kR = 3$ (Table 4). The restart itself released 5.7 dB on the 120° cap, and the further 25.5 dB on the 30° cap at $kR = 1$ lie below what the reference resolves, as shown below. Meeting the stopping rule does not, however, mean finding a local optimum. The rule fires once the scaled objective changes by less than about 0.04 dB per 100 iterations, whereas the exact runs from the sinc design were still descending by 0.07 to 0.12 dB per 100 iterations at iteration 3000; on the 90° and 120° caps they passed the runs from uniform that had stopped, by 2.8 and 1.8 dB, as panel (b) shows for the 90° cap. Here,

Table 4: Runs continued beyond the 600-iteration limit. Stop: iteration at which the exact run from the uniform design met the stopping rule; gain: additional level reduction from iteration 600 to the stopping iteration; sinc e: true minus predicted level of the sinc run from the uniform design, Eq. (8), at iterations 600 and 3000; sinc s: its level above the best exact design found in the case by iteration 3000, Eq. (10), a lower bound; spread: level difference between the exact runs from the uniform design and from the sinc design, at iterations 600 and 3000. Differences in dB. Cases in the order of Table 3; the design point had one exact run continued.

θ_0	kR	stop	gain	sinc e, 600 $\to$ 3000	sinc s	spread, 600 $\to$ 3000
30°	1	1618	25.5[a]		pending[b]	
90°	1.5	944	0.71	19.0 $\to$ 22.1	33.6	0.4 $\to$ 2.8
120°	1.5	1903	5.70	23.5 $\to$ 25.8	29.2	3.1 $\to$ 1.8
30°	3	1568	0.74	21.6 $\to$ 22.7	49.3[a]	5.8 $\to$ 2.7[a]
30°	18.3	4355	0.27	—	—	—

[a] Rests on exact designs below about −80 dB, which the reference does not resolve; with the factor $\ell_{\max} = 8000$ alone, the gap s at $kR = 3$ is about 45 dB.
[b] The stored XS/SU checkpoints reached the 3000-iteration budget but remain unaudited and are excluded; only the frozen 1800-iteration values are reported (Table 3).

"converged" means only that the stopping rule fired; it does not establish a local optimum. How long that takes depends on the depth of the cancellation being pursued: at low kR each outer iteration of the exact runs needed 5 to 7 inner GCMMA evaluations, against about 3 at the design point, and the levels kept falling for thousands of iterations.

The piston model does not catch up with more iterations. Over the 2400 added iterations its runs from uniform lowered their own predicted level by 2.8 to 4.3 dB but their true level by only 1.2 to 1.7 dB, so its error e grew in all three cases, and the gaps s of their designs remained at 29 to 49 dB. The reference resolves these sinc designs to within 0.5%, so this part of the result does not depend on its accuracy. The spread between the two exact runs of a case, by contrast, ended between 1.8 and 2.8 dB, against 0.4 to 5.8 dB at iteration 600. What the optimizer leaves on the table is thus small beside what the piston model leaves, and the line at about 10 dB of Section 4.2 separates the two with room to spare.

The longer runs also expose a limit of the reference. On the 30° cap at $kR = 1$ and 3, the deepest exact designs move by 4 to 12 dB when the extrapolated factor is replaced by its $\ell_{\max} = 8000$ part: the design returned by the exact run from uniform at $kR = 1$, at −94.9 dB, lies at −82.6 dB

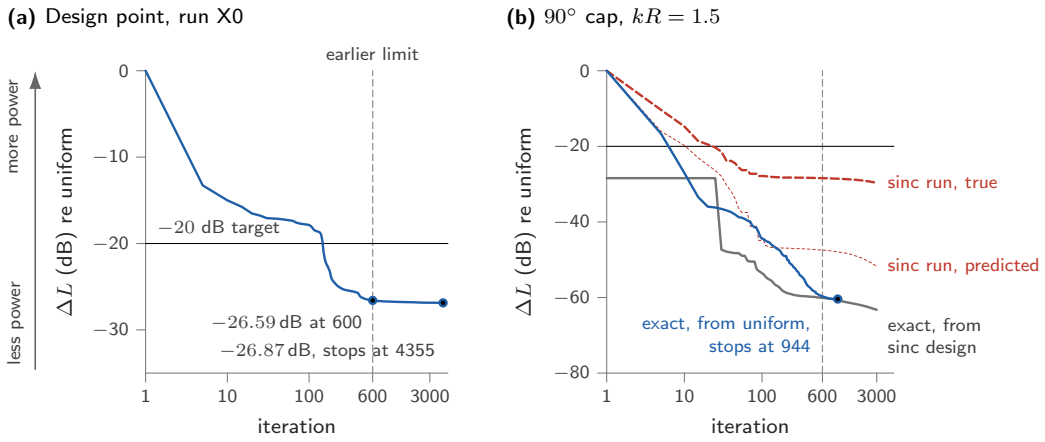


Figure 9: Level against iteration for runs continued beyond the 600-iteration limit; levels relative to the uniform design, higher means more power. (a) Design point: the exact-driven run X0, continued until it met the stopping rule. (b) 90° cap at $kR = 1.5$: true level of the exact runs from the uniform design (blue, solid; stopped by the rule at iteration 944) and from the end design of the sinc run (gray), and true (dashed) and predicted (dotted) level of the sinc run from the uniform design. Each curve follows the design the optimizer would return, the lowest value of its own objective so far. Iterations are counted from the start of each run, logarithmically; GCMMA was restarted at iteration 600 (vertical line). The horizontal line marks the −20 dB example target. Panel (b) is a scaled problem, with kR set through the sound speed.

under $\ell_{\max} = 8000$ alone, no lower than that of the run from the sinc design at iteration 1800, −83.1 dB. Run long enough, the optimizer exploits the residual error of the reference, so in these two cases depths below about −80 dB are unconfirmed. The returned designs of the 90° and 120° caps and of the design point agree between the two factors to within 0.05 dB.

To test whether a cheaper model can save iterations, Table J.1 compares six strategies on the 30° cap at $kR = 1$ and 3. Each started from the uniform design, used a 1200-iteration budget, and was fixed before results were read. At $kR = 3$, plain exact optimization led at every budget. At $kR = 1$, four staged runs reached its iteration-600 level after 349 to 474 iterations, but the exact run continued lower and ended lowest, even under the factor $\ell_{\max} = 8000$. Structural solves dominated loop time. Measured times exclude one-time operator construction (seconds for sinc and 70 to 250 s for an exact factor), so they do not establish a full setup-plus-run time advantage. At equal budgets of 900 and 1200 iterations, no staged strategy produced a lower exact level than plain exact optimization.

These results carry limits of their own. Each strategy ran once, from one start, in two cases, and GCMMA paths are sensitive to small perturbations (Appendix G), so differences of a few decibels lie within the spread of the optimizer. The switching rule reached deeper levels only on designs that violated the volume constraint, by up to 0.34%, and those levels were not counted. No independent reference, with a larger $\ell_{\max}$, a finer surface quadrature or more Galerkin functions, was run, and the cases at $kR = 7$ were not continued.

5. Robust design

The designs that reach the lowest levels are fragile. Across the scaled cases the levels of the exact optima rise by 2.2 to 14.5 dB when the added mass of the fluid, which carries 2×10^{-4} to 7×10^{-4} of the kinetic energy of the shell, enters the structural equations (Section 6). We examine independent Gaussian relative thickness errors with $\sigma = 1\%$, equal to 0.5 mm for a 50 mm plate; they are untruncated, so perturbed thicknesses may lie outside the nominal 25–100 mm design bounds. A design that reaches the target only at the nominal thickness has not reached it. The nominal-path envelope is not calibrated on robust-run iterates and is not a robust-design guarantee; robust performance is assessed separately on fresh thickness fields.

We therefore minimize the expected power of Eq. (6), with $\sigma = 0.01$ and an independent error for each of the 960 elements. Independent errors break the phase relations on which cancellation relies; spatially correlated errors [17] and eroded and dilated designs [16] are smoother models that we do not pursue. The expected value replaces the nominal objective as the expected response over an uncertain excitation frequency did in the work of Valentini, O. M. Silva and Cardoso [18]. The N samples are drawn once (seed 4242) and kept for the whole run, so GCMMA and its stopping rule are used unchanged (Algorithm 2). Each perturbed state and adjoint system is solved by GMRES preconditioned with the LU factors of the nominal matrix; at $\sigma = 0.01$ it converges in 6 iterations, and per sample it matches a direct solve to 3.5×10^{-10} in power and 1.6×10^{-9} in gradient. With $\sigma = 0$ the robust code reproduces the nominal objective to 6×10^{-14} and its gradient to 5×10^{-10}; at $\sigma = 0.01$ the adjoint gradient agrees with central differences to 3×10^{-7} to 7×10^{-6}, the level of the same test on the nominal objective (1.4×10^{-5}). The number of samples follows from the spread of the sample powers. For the design point, eight samples give a standard error of at most

0.35% of the expected power (0.015 dB), so $N = 8$ is used there; for the deep exact designs at $kR = 3$ the standard error is 21, 15 and 11% at $N = 8$, 16 and 32, and the three were compared.

To judge every design out of sample, we evaluated it on the same 200 fresh thickness fields (seed 7777) at $\sigma = 0.1$, 0.3, 1 and 3% (Table 5). The level of mean true power is $\Delta L_{x,\mathrm{mean}\,P} = 10\log_{10}(\mathrm{E}_\xi[P_{2s}]/P_{2s}(d_0))$. The entry labeled "worst 5%" is the 95th percentile of individual sample levels, not the expectation of decibels.

At the design point, the nominal designs are not fragile: X0 goes from −26.59 dB nominal to −26.54 dB expected at $\sigma = 1\%$ (95th percentile: −26.49 dB) and to −26.14 dB at 3%, while S0 stays at −17.09 dB.

At $kR = 3$, XU loses its cancellation. Its nominal level of −84.5 dB becomes an expected level of mean power of −65.3 dB at $\sigma = 1\%$; the 95th percentile of individual levels is −62.0 dB. At 3%, the expected level is −55.8 dB. The roughly 10 dB rise per factor of three in σ is consistent with power proportional to σ^2. Other deep exact designs in this case (XS, the 1568-iteration continuation of XU and the fluid-loaded re-optimization) end between −62.5 and −65.4 dB at $\sigma = 1\%$; a nominal depth beyond about −65 dB is of little use under this tolerance.

The sinc designs, near −34 dB, change by at most 0.03 dB. At $\sigma = 1\%$, XU remains 31.0 dB below SU in expected level; the gaps are 21.6 dB at 3% and 20.8 dB with fluid loading at the real air density (−53.9 against −33.2 dB), not the scaled medium of the last column of Table 5. A likely explanation is that the exact designs rely on cancellation between radiating and absorbing areas (Fig. 6), which random thickness errors perturb. We did not test that explanation at $kR = 3$, where the sinc designs lie far above the cancellation floor.

Optimization under uncertainty reproduces the nominal findings at the design point. From the uniform design the exact-robust run (800 iterations) ends at −26.53 dB expected and the sinc-robust run (800 iterations) at −17.43 dB; started from X0 and S0, they stay at −26.71 and −17.09 dB. In the low-frequency case ($kR = 3$), the exact-robust runs from XU (200 iterations) lowered their own objective on the eight, 16 or 32 training samples by 2.9, 2.0 and 1.0 dB, but on the fresh samples they ended at −64.3, −64.5 and −65.1 dB, against −65.3 dB for the nominal optimum XU: the optimizer fits the fixed samples, and the effect shrinks as N grows. The sinc-robust run from the uniform design (stopped at 300 iterations) reached −38.3 dB expected (worst 5%: −37.8 dB), 4.0 dB below the nominal sinc design SU.

Table 5: Level of mean true power $\Delta L_{x,\mathrm{mean}\,P}$ (in dB, relative to the nominal uniform design) under independent element thickness errors of standard deviation σ, evaluated on 200 fresh samples (seed 7777) for the 30° cap at $kR = 3$ and at the design point ($kR = 18.3$). Fluid loading at $\sigma = 0.01$ uses the scaled medium ($\alpha_f = 1$; at the design point, steel in air). Robust exact designs use $N = 32$ at $kR = 3$ (warm-started from the nominal exact optimum, 200 iterations) and $N = 8$ at $kR = 18.3$ (from the uniform start, 800 iterations); robust sinc designs use $N = 8$ from the uniform start (300 and 800 iterations).

		in vacuo, σ				$\sigma = 0.01$
Design	Nominal	0.001	0.003	0.01	0.03	fluid
30° cap, $kR = 3$						
exact optimum (XU)	−84.5[a]	−81.9[a]	−75.3	−65.3	−55.8	−63.9
sinc design (SU)	−34.3	−34.3	−34.3	−34.3	−34.2	−34.1
robust exact ($N = 32$)	−75.9	−75.4	−72.9	−65.1	−55.9	−63.1
robust sinc ($N = 8$)	−38.3	−38.3	−38.3	−38.3	−38.1	−37.9
30° cap, $kR = 18.3$ (design point)						
exact design (X0)	−26.6	−26.6	−26.6	−26.5	−26.1	−26.5
sinc design (S0)	−17.1	−17.1	−17.1	−17.1	−17.1	−17.0
robust exact ($N = 8$)	−26.6	−26.6	−26.6	−26.5	−26.1	−26.5
robust sinc ($N = 8$)	−17.4	−17.4	−17.4	−17.4	−17.4	−17.4

[a] Nominal and $\sigma = 0.001$ values for XU lie near the reference resolution floor ($\ell_{\max} = 8000$ gives −80.6 dB nominal); for $\sigma \geq 0.003$ all sample means lie well above that floor and agree between factors to within 0.1 dB.

Its predicted expected level is −51.9 dB; applying the nominal-path low-kR envelope of Section 4.3 gives −24.2 dB. That envelope would classify the robust design as passing the −20 dB target, conditionally; it was not calibrated on robust-run iterates, so the independent 200-sample evaluation, not the envelope, is evidence of its performance.

Robustness does not make the piston model enough. At $kR = 18.3$ the sinc-robust design lies 9.1 dB above the exact-robust design (−17.43 against −26.53 dB expected, and the same gap for the worst 5%) and misses the −20 dB target, which the exact-robust design meets. When $kR = 3$ the gap is 26.8 dB in expectation (−38.3 against −65.1 dB) and 24.1 dB for the worst 5%. The tolerance limits how deep the exact designs go, but it does not close the gap between the two models. Several runs were not made: no exact-robust run from the uniform design at $kR = 3$; two sinc-robust runs at

$kR = 3$ were stopped at 300 and 100 iterations because the shared machine was fully loaded; a mean-plus-standard-deviation objective was only checked for its gradient; fluid loading was not used inside a robust run; and only uncorrelated element errors were tested.

6. Fluid loading

Every power above was computed for a velocity set by the structure alone, which is a fair simplification for steel in air. The deep designs at low kR strain it: their exact efficiencies are 3×10^{-8} to 7×10^{-6} times that of the uniform design in the same case, so only a small part of their velocity escapes cancellation. We therefore coupled the shell to the fluid and evaluated the end designs again. The fluid-loading operator comes from the same exact solver: the virtual work of the pressure jump on the spline displacements yields a complex symmetric matrix $\mathbf{Z}$, whose resistive part carries the radiated power and whose reactive part acts mostly as an added mass. The coupled response solves

$$(\mathbf{K}_D + i\omega\,\mathbf{Z})\,\mathbf{q} = \mathbf{f} \tag{11}$$

with $\mathbf{K}_D$ of Eq. (1) unchanged; four checks of $\mathbf{Z}$, two independent implementations and a power balance support it (Appendix K). For the scaled cases, which fluid loads the shell needs a decision. The *scaled medium*, $\rho' = \rho_0 c_0 / c'$, keeps the impedance ratio of steel in air; the *air density* ρ_0 with the same c' keeps the mass ratio and raises the impedance by up to 18.3 at $kR = 1$. Neither is a physical heavy fluid, so the two readings bracket two ways of scaling the fluid, and the second is best read as a bound; at the design point they coincide.

For the four cases at $kR \le 3$, fluid loading raises the exact optima's radiated power by 20 to 30 dB with air density. The sinc designs shift by at most 0.25 dB in the scaled medium and 1.1 to 4.8 dB with air density; all remain below the -20 dB target. Figure 10 shows the casewise levels in both media.

The sinc, ERP and uniform designs barely notice the fluid: in the scaled medium their powers change by factors of 0.98 to 1.15. Their levels therefore survive. The sinc-driven designs of the six scaled cases stay at true levels of -26.7 to -41.8 dB in the scaled medium and -22.0 to -39.9 dB with the air density, below the -20 dB target in every case and both media. The exact optima at $kR \le 3$ respond otherwise: their powers grow by factors of 1.1 to

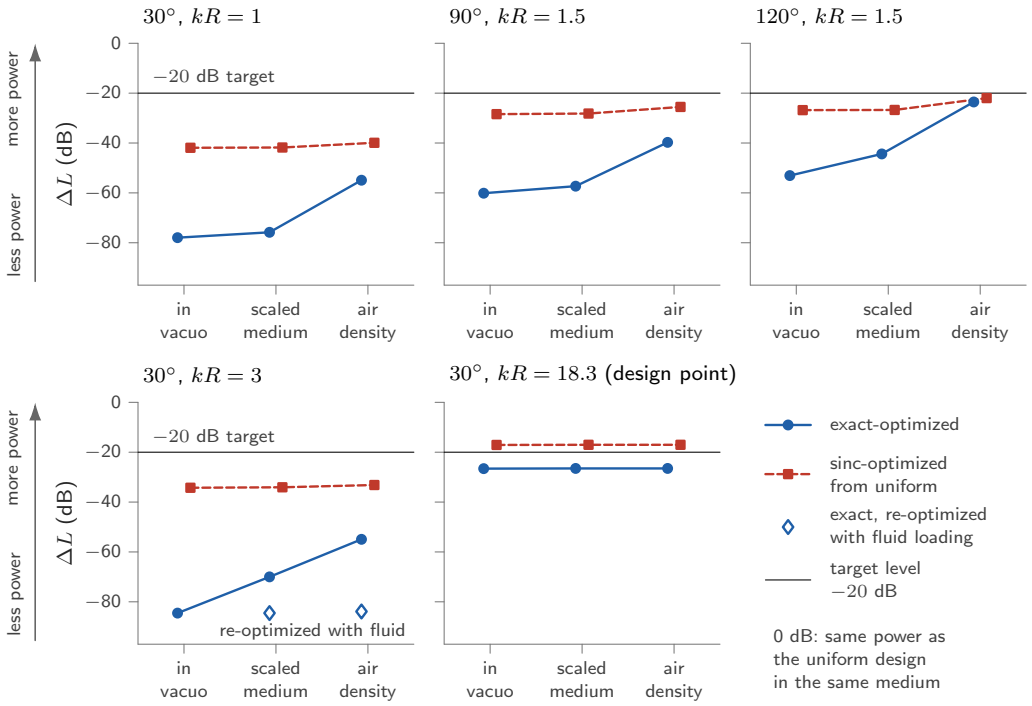


Figure 10: Fluid loading of the regime-study designs, one panel per case. Each panel shows the true level ΔL relative to the uniform design in the same medium: in vacuo, the scaled medium and air-density fluid. Zero dB marks the uniform design; the horizontal line marks the −20 dB example target. Blue solid: the better of the two exact-optimized designs in each medium; red dashed: the sinc design optimized from the uniform start. Open diamonds show the fluid-loaded exact re-optimizations for 30°, $kR = 3$. The gap s is the vertical distance between the blue and red lines; Table K.1 gives its values, the gaps to re-optimized designs and the two $kR = 7$ cases. Designs are optimized in vacuo unless marked. Zero-thickness two-sided model, structure driven at 100 Hz; horizontal shifts keep coincident points visible.

29 in the scaled medium and 35 to about 10^3 with air density. The largest change is at the 30° cap at $kR = 3$, where the reference leaves the factor uncertain by about three (Section 8); the corresponding level rises are 2.2 to 14.5 dB and 20 to 30 dB, respectively. At $kR = 7$ the best exact levels rise by 1.2 to 1.4 dB and 4.9 to 5.4 dB; at the design point, no design changes by more than 0.1 dB. The lowest 40 natural frequencies shift by at most 1.0%, well below a half-power bandwidth of about 10%, so detuning alone does not explain the rise. One plausible interpretation is that fluid loading changes field amplitude and phase by an amount comparable to the residual field

after cancellation. Such a perturbation should affect a cancellation design more than a design whose radiation scales with velocity.

Because the exact designs rise in level and the sinc designs hardly do, the gap, the vertical distance between the two lines of Fig. 10, shrinks. In the scaled medium it stays at 17.6 to 35.9 dB in the four low-kR cases; with the air density it falls to 15.0 and 21.8 dB for the 30° cap, 14.2 dB for the 90° cap and 1.6 dB for the 120° cap. To separate properties of the in-vacuo designs from those of the design space, we re-optimized the 30° cap at $kR = 3$ with fluid loading, one run per medium from the in-vacuo optimum. The calculated levels were −84.5 dB in the scaled medium and −83.9 dB with air density after 150 iterations, but both are below the reference floor (about −80 dB with the uncorrected $\ell_{\max} = 8000$ operator). They support only the conclusion that these designs reach at least about 80 dB below the uniform design, not the exact reported levels. Evaluated in vacuo, their levels rise by 14.6 and 29.7 dB, respectively. For this zero-thickness cap model and in the cases tested, the advantage of the exact operator at low kR survives only when the fluid loading is part of the model being optimized, and a design tuned to −80 dB in one model cannot be relied on in another. For the −20 dB target, the piston-driven designs were the steadier choice.

7. Reproducing the results

The model code, the scripts behind every stored result, the stored text results and the figure sources are in the repository `https://github.com/dshguilherme/piston`; the binary designs, operator factors and states (2.9 GB) and the longest run logs are listed there with their checksums and go to a separate dataset (see Data availability). Table A.2 gives the command behind each figure and table and its run time.

The structural model, optimizer and coupled re-optimization ran in MATLAB R2025a with GeoPDEs 3.2.2 [25], NURBS toolbox 1.4.3, and four first-author element routines. The MATLAB run logs do not record the installed release, and the GeoPDEs source came from a zip snapshot without a commit hash. The exact-cap solver, operator factors, fluid-loading operator and statistics used Python 3.13.12 (NumPy 2.5.2, SciPy 1.18.1, Matplotlib 3.11.2); the boundary-element checks used Python 3.11.17 (bempp-cl 0.4.2 [26], NumPy 2.4.6, SciPy 1.17.1, Numba 0.68.0). These versions come from freezes made when the repository was assembled, not from run logs; a prior local check reproduced the stored analytic-check logs byte for byte in the first

Python environment. Figures use TeX Live 2025 (pdfTeX, PGFPlots 1.18). All runs used one Windows 11 laptop (Intel i5-12500H, 12 cores, 15.7 GB) shared with other jobs, so reported times are upper bounds; memory use was not recorded.

The package works at three levels. Re-plotting rebuilds the 17 figures and the table cells from the stored results in a few minutes. The smoke tests add the analytic and operator checks; on a fresh copy of the repository they took about 2 min, the 23 plotted data files came out byte-identical to the stored ones and every computed table cell matched the paper, while the check of the exact operator on stored designs needs the binary dataset. A full re-run takes 13 to 15 h of MATLAB and Python and about 6 h of boundary-element solves; it needs MATLAB with GeoPDEs, the four element routines (included in the repository), both Python environments, about 16 GB of memory and the binary dataset. Re-evaluating stored designs without re-optimizing is possible with the same scripts, but it is not wrapped in a single command. The longest single steps are these. The exact factor of one case took 70 to 250 s to build (54 s at the design point), one optimization run of 600 iterations 2 to 18 min, and a re-optimization with fluid loading in the loop, with a dense factorization, about 14 s per iteration; no step needs a boundary-element solution except the verification of Appendix C.

Every random quantity has a fixed seed: the 40 random thickness fields of each cap (MATLAB, seed $1000+\theta_0$ in degrees), the random pattern combinations of Appendix E (NumPy, seed 20261016), the directions of the gradient checks (seed 517) and the thickness samples of the robust objective (seed 4242; seed 7777 for the evaluation samples). The thread count is part of the computation, too (Table A.1): the design-point runs used the MATLAB default, the regime-study and robust runs two threads, the long-horizon and fluid-loading re-optimization runs four threads, and the Python fluid-loading scripts four BLAS threads. A sparse factorization changes in the last digits with the thread count, about 3×10^{-10} in the states, and a replicate whose objective differed by a factor of $1+4\times10^{-8}$ drifted by up to 1% in power after 10 iterations (Appendix G). A run of 600 iterations is therefore not bitwise portable: on another machine, the verdict of each case is what a re-run can be expected to reproduce, rather than the last digits of a trajectory.

8. Discussion, limits and conclusions

We studied steel spherical caps of 10 m radius and 30°, 90° and 120° half-angle: a 30° design point in air at 100 Hz and six regime cases that vary the opening angle and an effective sound speed at the same drive. The main finding is that the accuracy of a model at its end designs does not decide whether those designs reach the target, because the optimizer samples the model error along its own path. At the design point the piston model predicted the levels of its own designs to a fraction of a decibel, yet its descent from the uniform start stopped short of the example target of −20 dB, which the exact-driven descent passed by more than 6 dB; the difference lay in the optimization trajectory rather than in the end-point rating. In the scaled cases at low kR the piston model placed its designs tens of decibels too low, yet their true levels stayed below −25 dB, and when the in-vacuo-optimized designs were reevaluated with fluid loading, they remained below the −20 dB target in both media. A simplified model can thus be inaccurate and still sufficient, or accurate and yet insufficient from a cold start. Running the optimizations for thousands of iterations instead of 600 changed none of these verdicts in the cases continued: the runs slowed down rather than converged, the error of the piston model grew slightly, and no strategy that let a cheaper model start the descent reached the exact level in fewer iterations, since every model costs the same per iteration.

The size of the error is why predictions alone cannot be trusted: at the −40 dB target, the piston model would have declared success in all six scaled cases, against one true success. The rule of Section 3.2 therefore adds a measured error budget to the predicted level. It is stated for any target T, and the 20 and 40 dB used here are arbitrary examples, but its empirical guarantee is conditional on the set of designs on which the budget was measured, since the ratio of piston to exact power varies over orders of magnitude between designs. With the budget fixed before the outcomes were read, from piston-driven designs of the same kR class and allowed to grow as the predicted level falls, the empirical envelope covered all 14 tested piston-driven end designs in vacuo, accepted all 13 that reached the −20 dB target, and gave no false accept at the −40 dB target in this set. A single largest error over random or exact-driven designs is not a valid bound, since those designs never reach the low predicted levels where the error of the piston model is largest. At $kR \leq 3$, the piston model supports the −20 dB target only when it predicts about −44 dB or lower, and it cannot certify the −40 dB target;

at $kR \geq 7$ and at the design point, the required margin for $T = 20$ dB is about 1 dB. These statements rest on three or four calibration cases per class, two sinc runs per case, and the in-vacuo truth. Fluid-loaded results above are reevaluations of those in-vacuo designs, not re-optimizations; the bound held for half the designs at air density. Only the exact 30° cap at $kR = 3$ was re-optimized with fluid loading, one run per medium, and both reported depths lie below the reference floor.

Three of the 14 predictions extrapolate beyond the fitted range, and all designs come from correlated paths in this cap family; another geometry, frequency, load or support needs its own calibration set and held-out test against its own reference.

At low kR, the exact optimizer finds designs from -53 to at least about -80 dB, far below the piston-driven designs. Levels near -80 dB sit at the floor of the reference, lose part of their depth when the fluid loads the shell, and are re-found only by an optimizer that includes the fluid. Such levels rely on sensitive physical and mathematical properties, which are very difficult to manufacture and to observe in reality. A target met by cancellation must also tolerate manufacturing scatter. Under independent 1% element thickness errors (Section 5) the exact optimum of the 30° cap at $kR = 3$ rose from -84.5 to -65.3 dB in expected level, and every deep exact design of that case ended near -65 dB; at the design point the deepest exact design lost 0.05 dB. Designing for the expected power did not make the piston model enough. Its robust design stayed 9.1 dB above the exact-robust design at the design point, where it misses the -20 dB target, and 26.8 dB above it at $kR = 3$, where both meet that target. The tolerance limits how deep the exact designs go, but it does not close the gap between the models.

These findings carry limits. Every two-sided number rests on the idealization of a surface of zero thickness, one normal velocity for both faces and no cavity, so that "exact" means exact for a prescribed velocity, or, in Section 6, for the coupled problem with the radiating surface at the mid-surface. The thickness is therefore treated in two different ways. The structural model represents it within Kirchhoff–Love theory: it is the design variable ($25 \leq h_e \leq 100$ mm on a cap of radius 10 m, so $h/R \leq 0.01$) and the random variable of the robust objective, so its effect on mass and stiffness is included. The acoustic model leaves it out: the two faces of the real shell lie up to 50 mm on either side of the mid-surface, and the radiation from the rim thickness and from the offset between the faces was not modeled. Both effects are small against the wavelength of 3.4 m, but we did not quantify

them. The structural fields come from a pinned rim assumed acoustically transparent. The design-point study covers one geometry at one frequency with a local gradient method, three starts per operator that are not independent, and an exact run, X0, that had not converged, so its gaps are lower bounds. The regime study adds a scaled problem with kR set through an effective sound speed, one load and one drive frequency; one start per objective from the uniform design plus cross-starts; 34 of 42 runs at the iteration limit and near-singular GCMMA subproblems in five runs at $kR \leq 1.5$; and a gap threshold set after the data were in. Continued for up to 4355 iterations in five cases, the runs slowed down without a verified local optimum, and the comparison of staged strategies rests on one start per strategy in two cases (Section 4.4). On the 90° and 120° caps the reference was not re-checked independently, and the structural mesh was not refined: at the equator its elements are about 2.6 m long around the axis, against a bending wavelength of about 2.2 m at 100 Hz, so those designs are not converged physical designs. The reference resolves the end designs to within 5%, except the exact designs of the 30° cap at $kR = 3$ (about ±2 dB in vacuo). Run longer, the exact optimizer exploits the residual error of the reference: on the 30° cap at $kR = 1$ and 3 its deepest designs move by 4 to 12 dB when the extrapolated factor is replaced by its $\ell_{\max} = 8000$ part, so depths below about −80 dB in these two cases are unconfirmed, and no independent reference with a larger $\ell_{\max}$, a finer quadrature or more Galerkin functions was run. With fluid loading the deepest powers are resolved to about 30%, and below about 10^{-10} of the input power the power balance cannot confirm them. The fluid-loading check evaluated end designs only, froze the added mass at 100 Hz for the modal shifts, omitted azimuthal orders above 20 (less than 0.14% of the surface integral of $|v_n|^2$), re-optimized one case with one start, and did not re-optimize the sinc objective with the fluid. The tolerance study of Section 5 assumes independent element errors of the thickness only and uses 8 to 32 fixed samples, which the exact optimizer fits at $kR = 3$; it has no exact-robust run from the uniform design at $kR = 3$, and its two sinc-robust runs there were stopped early.

The limits point to the open questions. Where between $kR = 3$ and 7 the piston model stops placing its designs tens of decibels too low is not located, since no case lies there; cases at $kR = 4$ and 5, several starts per objective, and an independent re-check of the deepest exact designs would sharpen the picture. Shells without an exact solution, such as the cylindrical panel of Appendix B, need a two-sided boundary-element operator reduced

to the structural coefficients in the form of $\mathbf{F}_x$, built at a resolution that can be trusted for canceling fields; with it, a shell-specific error budget could be calibrated and tested on representative held-out designs. Independent geometries, operating-frequency sweeps, support conditions and fluid scalings are still needed to assess transfer across shell applications.

Appendix A. Settings, numerical details and reproduction

Table A.1 completes Table 1 with the settings of the operators, the objectives and the optimizer, in the order of the chain of Fig. 2.

The adjoint gradients of Section 2.5 were checked against central differences in a random direction drawn with a fixed seed, taking the best of three steps at the design point. The 40 random smooth thickness fields of each case of the regime study are drawn with a fixed seed per cap as cosine and sine series in the polar and azimuthal coordinates (orders 0 to 3, weights decreasing with the order) with amplitudes of 5 to 25 mm, projected onto the bounds at a mean of 50 mm. The evaluation of Algorithm 3 uses the sinc and exact operators at the design point, without ERP and without the resolution check with 32 Galerkin functions, both of which are added in the regime study.

Some steps cannot be repeated exactly from the repository. The original scripts still carry absolute paths of the authors' machine (16 MATLAB and 12 Python scripts) and write into `results/`; `config.m` puts the right folders on the path, but their outputs stay away from the stored results only if they are run on a copy of the repository. The four GeoPDEs element routines are included in `vendor/`; GeoPDEs and the NURBS toolbox themselves must be downloaded. Two stored logs cannot be regenerated verbatim: the boundary-element validation log, whose sign line no longer matches the current code, and the flat-limit table, never written because the case $a/R = 0.05$ ran out of memory (the log of the other cases is kept). The lists of runs passed to the regime-study queue, the call lines of the two re-optimizations with fluid loading and the paths of their impedance files were not recorded; the queue list in `drivers/` is a reconstruction from the queue log. The frozen designs of Appendix D and the panel field of Appendix B are inputs; their generators were recovered from the version history, but the design file predates later changes to the model code, so a re-run is not guaranteed to reproduce it byte for byte. Every point of Fig. 10 is read from the `.dat` tables in `figs/data/`, none is typed in its source, and `make_tab_fsi.py` checks Table K.1 against

Table A.1: Settings of the operators, objectives and optimizer (the model is in Table 1).

Item	Setting
ERP operator	$\frac{1}{2}\rho_0 c_0 \int \lvert v_n\rvert^2 dS$, 8×8 Gauss points per element
Sinc operator	spline-consistent, 4×4 Gauss points per element (15 360 points); j_0 written as a plane-wave average and integrated over 32 Gauss–Legendre nodes in $\cos\vartheta$ times 64 equally spaced azimuths, a 2048×3024 factor (Appendix B)
Exact operator	$\mathbf{F}_x$: per azimuthal order $\lvert m\rvert \leq 20$, 48 Galerkin functions, of which 35 to 48 remain after whitening (Gram eigenvalues below 10^{-12} of the largest dropped), 1730 rows in all at $\theta_0 = 30°$ (1796 at 90°, 1834 at 120°); velocity at 8×8 Gauss points per element; Richardson step $2\mathbf{H}_{8000} - \mathbf{H}_{4000}$; negative eigenvalues set to zero; block form $\frac{1}{4}\rho_0 c_0(\mathbf{A}^{-1} + \mathbf{A}^{-\mathrm{H}})$ for the design-point runs, $\frac{1}{2}\rho_0 c_0\, \mathbf{A}^{-1}(\mathrm{Re}\,\mathbf{A})\mathbf{A}^{-\mathrm{H}}$ in the regime study and with fluid loading (Appendix C)
Fluid loading	$\mathbf{Z}$ from the same solver and basis: resistive part from the second block form, reactive part from $\rho_0 c_0\, \mathbf{A}^{-1}$, Richardson step, conjugated to $e^{i\omega t}$, symmetrized for the re-optimization; scaled medium $\rho' = \rho_0 c_0/c'$ or air density ρ_0
Objective	Eq. (5), $A = 100$, J in W; regime study, re-optimization with fluid loading and robust runs: operator scaled so that $J = J_{\mathrm{ref}} = 5.9464 \times 10^{-7}$ W at the start design, uniform or warm; design-point runs unscaled (Section 2.4)
Robust objective	expected power under independent element errors $h_e(1 + \sigma\xi_e)$, ξ_e standard normal, $\sigma = 0.01$; sample average over $N = 8$ samples ($N \in \{8, 16, 32\}$ in Section 5), drawn once from a fixed normal bank (seed 4242) and kept for the run; perturbed systems solved by GMRES preconditioned with the nominal LU (tolerance 10^{-10}, limit 60 iterations, direct-LU fallback); evaluated on 200 fresh fields (seed 7777)
Optimizer	GCMMA [24]; at most 600 outer iterations (150 for the re-optimization with fluid loading, 5000 in Section 4.4). Changed from the defaults of the implementation: at most 20 inner iterations (default 10), conservatism growth factor 1.3 (1.1), lower bounds of the conservatism parameters 10^{-3} for objective and constraint (10^{-6}). Defaults kept: initial asymptotes at 0.5 of the bound range from the design, asymptote factors 0.7 and 1.2, asymptote distances between 0.01 and 10 bound ranges, subproblem box factor 0.1, subproblem tolerance 10^{-7}, $a_0 = 1$, $a = 0$, $c = 1000$, $d = 1$; no move limit
Stopping rule	checked from the 21st iteration; for three consecutive iterations: relative change below 10^{-4} between the mean scaled objectives of two successive 10-iteration windows, largest thickness change below 10^{-4} of the bound range, and $g \leq 10^{-6}$; the design returned is the iterate with the lowest J, not necessarily the last
Threads	MATLAB default (design point); 2 (regime study, Section 4.4, and robust runs, Section 5); 4 (re-optimization with fluid loading)

Table A.2: Re-plot commands and timings; the last column gives the full-run stages and their measured wall times. `mf` is short for `python reproduce/figures/make_figures.py --paper`; all commands run from the root of the repository. Times are from the authors' laptop, shared with other jobs.

Item	Re-plot command	Time	Full re-run
Fig. 1	`mf --only fig_cap_geometry`	15 s	model parameters only
Figs. 2, 3	`mf --only fig_pipeline_eng fig_cap_problem`	10 s each	schematics
Table 2, Fig. 4	`python reproduce/tables/make_tab_opt.py`; `mf --only fig_opt_paths_v2`	1 s, 10 s	factor $\mathbf{F}_x$ 54 s; six runs 66 min; path evaluation minutes
Figs. 5, 6	`mf --only fig_designs_paper_v2 fig_mechanism`	2 min	states of the six runs (dataset)
Table 3, Fig. 7	`python reproduce/tables/make_tab_pilot.py`; `mf --only fig_pilot_regime_v2`	1 s, 10 s	seven factors 15 min; the 36 regime runs 4.8 h (2 threads), the six design-point runs 1.0 h; evaluation and statistics minutes
Fig. 8, Table I.1, Fig. I.1	`python tests/r0_passrule_stats.py`; `python tests/r0_fig_passrule_v2.py`; `mf --only fig_passbars_v2 fig_passrule_scatter_v2`	under 1 min	from the regime-study evaluation
Fig. 10, Table K.1	`python tests/r0_fig_fsi_f18b.py`; `mf --only fig_fsi_v2`; `python reproduce/tables/make_tab_fsi.py`	10 s, 1 s	**Z** 25–38 s per case; coupled checks 31 min; two re-optimizations 34 and 33 min (4 threads)
Tables 1, A.1	`python reproduce/tables/make_tab_setup.py`	1 s	checks the settings against the code
Table 4, Fig. 9, Table J.1	`python tests/r0_long_summary.py`; `python tests/r0_fig_longruns.py`; `mf --only fig_longruns`	under 1 min	continued runs and strategies about 13 h (2 threads, `tests/r0_long_runs.m`)
Table 5 (Section 5)	`python tests/r0_robust_summary.py` on each `evalstudy` file	2 s	fragility and robust runs (`tests/r0_robust_runs.m`, `r0_robust_eval.m`, `r0_robust_evalstudy.m`; 2 threads)
Appendix figures	`mf`	3 min	exact-solver sweeps 2–9 min each; boundary-element reference about 6 h

the stored fluid-loading results. Lastly, the environment files list the versions that ran, but no fresh environment was built from them.

Appendix B. The sinc model: factor two, discrete matrix and compression

The coefficient α of Eq. (2) hides a factor two. Removing the baffle and adding a mirror image of the patch leaves the pressure above the plane unchanged. One free-space sheet of monopoles, with volume velocity v_n per unit area, radiates $P_{\mathrm{mono}} = \frac{1}{2} P_{\mathrm{sinc}}$ on any surface, because the imaginary part of

the free-space Green function $e^{ikr}/(4\pi r)$ (time factor $e^{-i\omega t}$) is $\sin(kr)/(4\pi r)$. We checked this on 300 random points of a non-spherical surface, with P_{mono} computed from its far field: the ratio was 2 to twelve digits at four wavenumbers. The identity follows from the monopole-element power model [27] together with image doubling for a baffle, and it pulls two questions apart: the sinc model is twice one free-space source layer, and whether that matches a real shell depends on how many faces radiate. A constant factor, such as $2\times$ sinc, changes ρ but neither the level error of Eq. (8) nor the minimizer, so doubling is a calibration of the sinc model rather than a distinct model.

The discrete sinc matrix depends on the velocity trace and on the surface integration. For design G at 100 Hz, an element-centroid matrix (one distance per pair of elements and a constant velocity per element), the kernel integrated over finite patches, and the spline-consistent matrix give 7.091, 7.142 and 7.185 nW. The spline-consistent matrix takes the field and the normal from the spline basis at every quadrature point $\mathbf{x}_q$, so that $P_{\mathrm{sinc}} = \omega^2 \mathbf{q}^H \mathbf{S} \mathbf{q}$ with $\mathbf{S} = \alpha\, \mathbf{B}^H \mathbf{W} \mathbf{K} \mathbf{W} \mathbf{B}$ and $K_{qr} = j_0(k|\mathbf{x}_q - \mathbf{x}_r|)$. The dense $15\,360 \times 15\,360$ kernel $\mathbf{K}$ is never formed. The code uses the plane-wave identity

$$j_0(k|\mathbf{x}-\mathbf{y}|) = \frac{1}{4\pi} \int_{S^2} e^{ik\mathbf{s}\cdot(\mathbf{x}-\mathbf{y})}\, d\Omega_{\mathbf{s}}, \tag{B.1}$$

integrated over the unit sphere of directions $\mathbf{s}$ with the 32×64 quadrature of Algorithm 1. This gives $\mathbf{S} \approx \mathbf{F}_{\mathrm{S}}^{\mathrm{H}} \mathbf{F}_{\mathrm{S}}$ with the 2048×3024 factor of Table A.1, positive semidefinite by construction and convergent with the number of directions, independently of the surface quadrature. Four checks support it. Tested on 13 random points at 100 Hz, the direction quadrature reproduced the dense matrix to 5.7×10^{-16} in relative Frobenius norm, and for a uniform piston in a baffle, with 48×96 directions, the closed-form power to 4×10^{-14} at $ka = 0.1$ to 6. Across the six frozen cap fields at 100 Hz, going from 24×48 to 64×128 directions changed the sinc power by less than 10^{-14} relative; of all cases in this paper, this one has the largest $k|\mathbf{x}-\mathbf{y}|$ on the surface, 18.3 (14 in the regime study). An independent evaluation of the sinc power from its spherical-harmonic (Funk–Hecke) series on the cap agrees to 10^{-7} for all 3121 designs of the design-point runs, and in the regime study to medians of 2×10^{-7} to 2.5×10^{-6} per case, with a largest difference of 1.5×10^{-4}, the series being stored there in single precision. It gives the sinc powers of this paper, with two exceptions. The face-power comparison of Appendix F uses the sinc matrix on the boundary-element mesh. Design G was produced earlier with the element-centroid matrix of the thesis [8] in

a low-rank (Nyström) form; started from design G, an optimizer driven by the spline-consistent matrix changed its sinc level by less than 0.001 dB (run SG). Going from the centroid matrix to the spline-consistent one changed the power by 1.33% for design G and by at most 1.99% over the six frozen cap fields, minor beside the model error of a factor 1.6 to 2.0 at 100 Hz.

Compression is another matter. A field of a cylindrical panel from separate optimization runs at 6.54 Hz, whose setup is not described in this paper, illustrates it; its low power results from cancellation. A rank-32 Nyström approximation of the kernel, with a relative kernel error of 1.18×10^{-5}, underestimated the power of this canceling field by 6.13%. A small normwise error need not give a small error in a small quadratic form, as for small eigenvalues computed with normwise-accurate methods [28], and weakly radiating modes are the most sensitive to perturbation of the radiation operator [3]. The boundary-element ratio of that panel is 0.437 for a uniform field and about 0.02 for the canceling field, the latter known only to an order of magnitude. The exact cap shows how far the sinc model can misjudge such a field. We searched an eight-pattern analytic subspace, which the reference resolves in 114 of the 294 cap and frequency cases. On nearly closed caps, the pattern in it that minimizes the sinc radiation efficiency reaches, at its smallest, $\rho = 0.048$ for $\theta_0 = 120^\circ$ and $\rho = 0.011$ for 150°, so in the worst case for each cap the sinc model rates it about 21 and 95 times quieter than it is.

Appendix C. The exact reference: verification and reduced operator

Two independent checks support the exact solution (Fig. C.1). As $a/R \to 0$ the cap tends to a flat disc in free space, solved separately in the Hankel-transform domain; the extrapolated cap power agrees with the disc at $ka = 0.1$, 1, 3 and 10 to the differences listed in the caption. A boundary-element solution with bempp-cl [26], using the Burton–Miller formulation [29], reproduces the exact breathing-sphere solution at $kR = 1$ to 1.4×10^{-4}. Both solvers were given the same piecewise-linear velocity trace on each boundary-element mesh for the six frozen cap fields at 100 Hz, and they agree to within 1.03% on the coarse mesh and 0.1–0.6% on the fine mesh (5.2 and 7.9 elements per wavelength). A second comparison uses the earlier boundary-element powers, extrapolated linearly in the element size from the two meshes, against the exact power for the same trace on a mesh refined to level 8 (the finest trace, which changes the sinc power by 0.04 to 0.09% relative to the spline

field). One caution concerns low frequencies: at $kR = 0.1$ the boundary-element surface power is wrong by a factor of up to 12, whereas its far-field power stays within 2.5% of the exact value at $kR = 0.1$, 0.3 and 1. We did not investigate the cause. The boundary-element points of Appendix D are therefore far-field powers; they follow the exact curve, with ratios between 0.97 and 1.007 over all frequencies, the largest differences occurring at $kR = 21$, where the mesh is under-resolved, and at $kR = 8$–10 for design G. Besides the basis size, the main approximation is the truncation degree $\ell_{\max}$: Richardson extrapolation from $\ell_{\max} = 4000$ and 8000 changes the power by 1.3 to 2.0×10^{-4} for five of the six frozen fields and by 6.7×10^{-4} for design B, and for all 3121 designs met along the six design-point runs dropping the Richardson step changes the exact power by at most 0.14%.

The reduced operator $\mathbf{F}_x$ is built from the same solver as in Algorithm 1; at $\theta_0 = 30^\circ$ the whitening keeps all 48 edge functions for $|m| \le 4$ and 35 at $|m| = 20$, 1730 in all, and each block $\mathbf{H}_m(\ell_{\max})$ is Hermitian and positive semidefinite in exact arithmetic. Two algebraically equal forms of the block were used. Because $\mathbf{A}_m$ is complex symmetric, $\mathbf{A}_m + \mathbf{A}_m^{\mathrm{H}} = 2\,\mathrm{Re}\,\mathbf{A}_m$, and

$$\tfrac{1}{4}\rho_0 c_0\,(\mathbf{A}_m^{-1} + \mathbf{A}_m^{-\mathrm{H}}) = \tfrac{1}{2}\rho_0 c_0\,\mathbf{A}_m^{-1}(\mathrm{Re}\,\mathbf{A}_m)\,\mathbf{A}_m^{-\mathrm{H}}, \tag{C.1}$$

where $\mathrm{Re}\,\mathbf{A}_m$ carries the real parts $\mathrm{Re}\,\lambda_\ell = x^2 j_\ell'(x)^2 \ge 0$, so the right-hand side is positive semidefinite term by term and involves no difference of nearly equal numbers. The design-point runs of Table 2 and Figs. 4 and 5 use the left-hand form; Fig. 6 solves the dual series directly for each design. The regime study, including the evaluation of all its designs in the design-point case, and the fluid-loading operator of Section 6 use the right-hand form, rebuilt for each cap and wavenumber. In the left-hand form the negative eigenvalues come from the extrapolation and, in the high orders, from round-off: for $|m| \le 12$ the largest in magnitude is 1.1×10^{-4} of the block maximum, but for $|m| \ge 17$ it equals the maximum in size; the blocks with $|m| \ge 13$ have maxima at least five orders of magnitude below those of the low orders. The right-hand form keeps the most negative eigenvalue at or below 5.5×10^{-4} of the block maximum in every order and case. Assembled over all orders before clipping, the two forms differ by 2.8×10^{-10} at the design point and by at most 1.4×10^{-7} (the 30° cap at $kR = 1$) in relative Frobenius norm. After clipping, the two factors give the same power to seven digits for the six frozen designs at the design point.

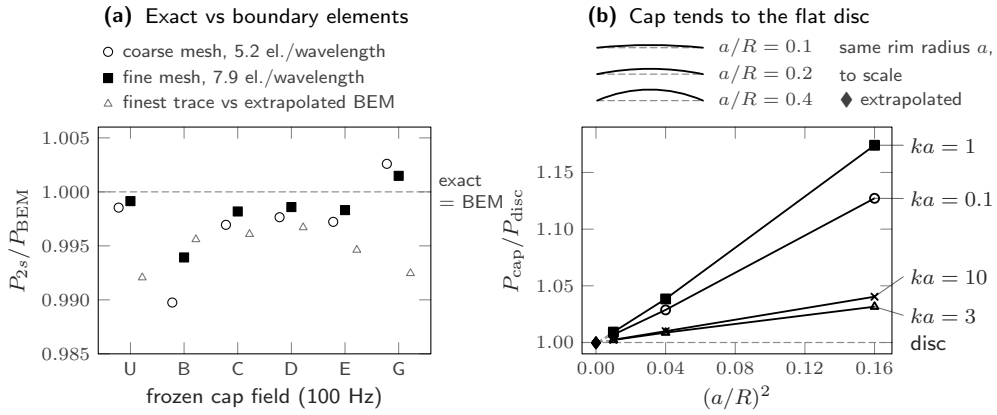


Figure C.1: Verification of the exact two-sided cap solution by two independent methods. (a) Exact power divided by boundary-element power for the six frozen cap fields at 100 Hz ($R = 10$ m, $\theta_0 = 30°$): U, the uniform-thickness design; B to E and G, the designs of Appendix D. On identical traces the ratio moves toward one for every field as the mesh is refined from 5.2 to 7.9 elements per wavelength; the finest trace against the extrapolated boundary-element power agrees to within 0.8%. (b) Uniform-cap power divided by the independent flat-disc power, $P_{\mathrm{cap}}/P_{\mathrm{disc}}$, against $(a/R)^2$ at $ka = 0.1$, 1, 3 and 10, for caps with $a/R = 0.1$, 0.2 and 0.4 at fixed rim radius a. The excess falls roughly in proportion to $(a/R)^2$, and the extrapolated values at $(a/R)^2 = 0$ (diamonds) differ from the disc by 3.9×10^{-4}, 5.4×10^{-4}, 3.2×10^{-5} and 1.6×10^{-4} at the four ka. Inset: the three caps drawn to scale.

Appendix D. Absolute power error on frozen fields

These tests hold a normal-velocity field fixed and change only the operator; we call such fields frozen. They test operators rather than optimized designs, since at another frequency the shell would vibrate differently and an optimizer would move elsewhere. We take six cap fields at 100 Hz from earlier optimization runs of ours with the same structural model, bounds and volume constraint, not published separately: the response of the uniform design (U) and of five optimized designs. Design G minimized the radiated power under the sinc model, in the element-centroid form of Appendix B, compressed; designs B to E minimized a net-volume-velocity measure (B), the input power (C), the dynamic compliance (D) and a phase-blind sum of squared velocity coefficients (E). Figure D.1 plots the ratio ρ for design G (bold) over $0.1 \le kR \le 40$. Figure D.2 extends the comparison to other opening angles and velocity patterns, and Fig. D.3 averages it over $10 \le kR \le 40$.

When the cap is acoustically small, ρ is far above one: 354 to 2627 at

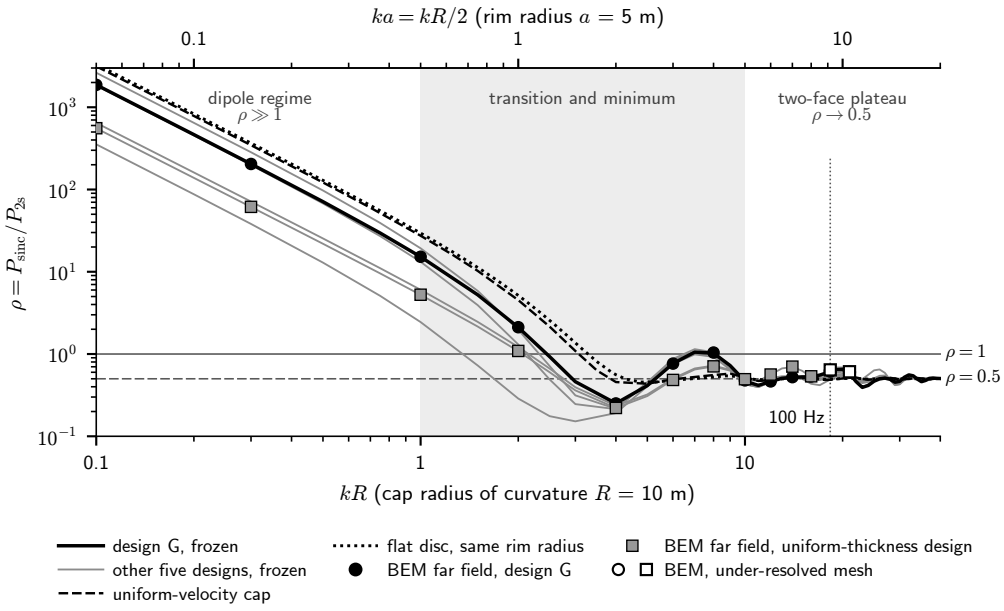


Figure D.1: Ratio of sinc power to exact two-sided power for six cap fields frozen at their 100 Hz shapes, against kR (bottom) and ka (top). From about 2×10^3 the ratio of design G falls through one to its minimum near $kR = 4$, rises back above one at $kR = 7$–8, and then oscillates about 0.5. A uniform-velocity cap (dashed) follows the flat disc of the same rim radius (dotted). Markers: spline-consistent sinc power over the boundary-element far-field power on the coarse mesh, for design G and the uniform-thickness design, open where the mesh has fewer than six elements per wavelength. The regime names and the shaded band are labels, not fitted limits. $R = 10$ m, $\theta_0 = 30°$; the dotted vertical line marks 100 Hz ($kR = 18.32$).

$kR = 0.1$, and 2.4 to 19 at $kR = 1$. The sinc model behaves like a monopole layer there, whereas the two faces of the open shell move together and radiate like a dipole layer. Between $kR \approx 1$ and 2.5 every ratio falls through one, and each field reaches its minimum, 0.15 to 0.25, at $kR = 3$ or 4; at $kR = 3$ alone the six ratios spread from 0.15 to 0.46, because the curves fall steeply there. Three fields rise back above one at $kR = 7$–8, to between 1.05 and 1.15, and a fourth reaches 0.99. Above $kR \approx 10$ the ratio oscillates about 0.5, within 0.39 to 0.70 up to $kR = 27$ and within 0.42 to 0.56 from $kR = 28$; at $kR = 40$ all six fields lie between 0.495 and 0.504. The flat disc gives the known baffled-versus-unbaffled behavior of flat plates [9, 10] and of rigid discs [30], and the uniform-velocity cap follows it closely (3147 against 3327

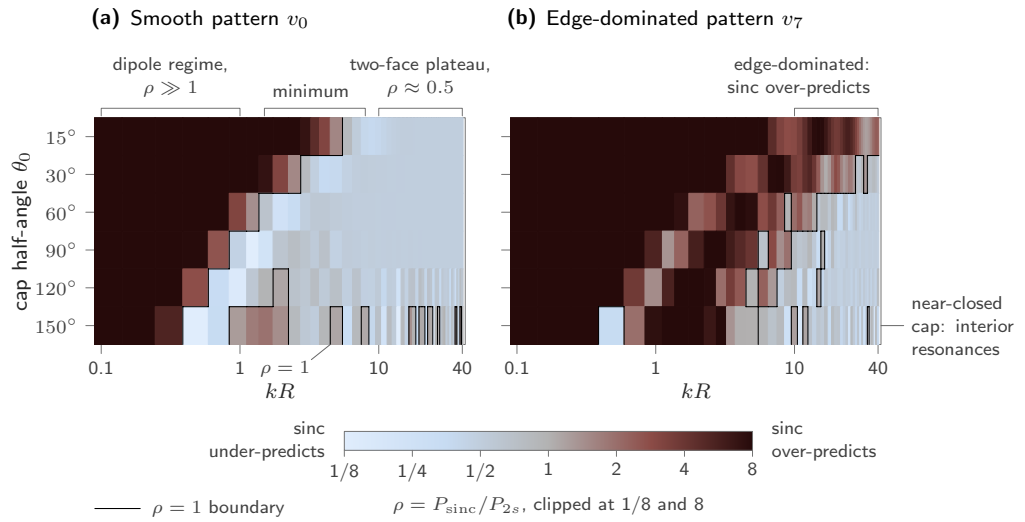


Figure D.2: Regime map of $\rho = P_{\mathrm{sinc}}/P_{2s}$ for open caps of half-angle θ_0, against kR. (a) Smooth pattern v_0; (b) edge-dominated pattern v_7. Darker cells mean that the sinc model over-predicts more, lighter cells that it under-predicts ($\rho = 1$ is mid-gray), and the black staircase separates $\rho > 1$ from $\rho < 1$. Each cell is one computed value, neither averaged nor interpolated; the shading is clipped at 1/8 and 8 (values at low kR reach 10^5). The regime names are descriptive.

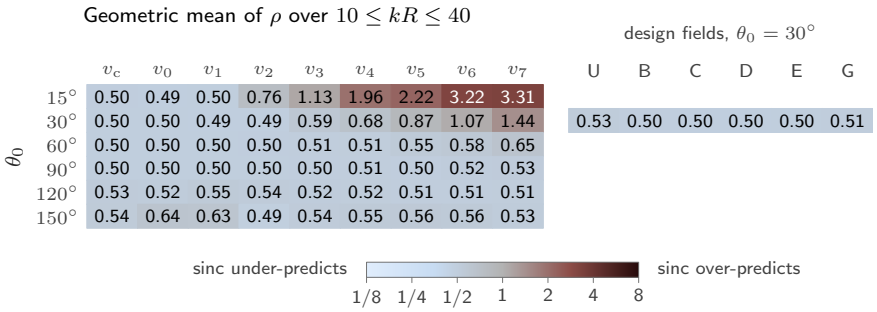


Figure D.3: Geometric mean of $\rho = P_{\mathrm{sinc}}/P_{2s}$ over $10 \le kR \le 40$: left, for the uniform velocity v_c and the patterns v_0 to v_7 at each half-angle θ_0 (31 sampled kR); right, for the six frozen design fields at $\theta_0 = 30°$ (32 sampled kR, including 100 Hz). Same shading as Fig. D.2: darker means that the sinc model over-predicts more.

at $kR = 0.1$, 0.47 to 0.53 above $kR = 10$). The six design fields follow its shape at a lower amplitude: for $kR \le 1$ they lie a factor of 1.3 to 11 below it, and near $kR = 8$ up to a factor of 2 above it, so the velocity pattern alone changes the ratio by a factor of up to 7.4 at $kR = 0.1$. At 100 Hz the six

frozen designs have ρ between 0.494 and 0.636; design G lies at -17.21 dB relative to the uniform design under the sinc model and at -17.04 dB under the exact model, and design B at -11.18 and -10.08 dB. Their level errors, in the sign convention of Eq. (8), are $+1.10$ dB for B, $+0.71$, $+0.82$ and $+0.54$ dB for C to E, and $+0.17$ dB for G: all five designs are louder than the sinc model predicts.

One size parameter does not place a case on the map. We evaluated caps with $\theta_0 = 15^\circ$ to 150° under a uniform velocity v_c and eight axisymmetric patterns $v_j = \cos\big((j + \frac{1}{2})\pi\theta/\theta_0\big)$, $j = 0, \ldots, 7$, which vanish at the edge and change sign more often as j grows (Fig. D.2). For open caps and smooth patterns the ratio stays close to 0.50 above $kR = 10$. When a small cap carries a rapidly varying pattern, the pattern is subsonic and the sinc model over-predicts: the geometric mean over $10 \le kR \le 40$ reaches 3.3 (Fig. D.3), and single values about 17, at $\theta_0 = 15^\circ$. On the nearly closed cap, $\theta_0 = 150^\circ$, the ratio oscillates between about 0.05 and 1.8. Three quantities were needed to place a case on the map: kR, θ_0, and how fast the velocity pattern varies compared with the acoustic wavelength. Doubling the sinc power calibrates the level only on the plateau: for the six frozen fields at $kR \ge 10$ it lands within -16% and $+22\%$ of the exact power (5th to 95th percentile), and for the analytic patterns at $kR \ge 10$ within $\pm 20\%$ for 92% of the points at $\theta_0 = 90^\circ$, but only 34% at 15° and 18% at 150°.

Appendix E. Rank agreement between the two operators

The agreement in order is measured by the Kendall rank correlation τ between sinc and exact power over the six frozen fields of Appendix D, which equals one when no pair is swapped. At 100 Hz, $\tau = 1$, and so it is at 29 of the 48 frequencies of the sweep. On the other 19, τ dropped to between 0.47 and 0.87: at every frequency up to $kR = 3$, at six between $kR = 4$ and 13, and at three between $kR = 20$ and 27. With six fields, τ moves in steps of $2/15$, so 0.87 means one swapped pair out of fifteen, and the lowest value, at $kR = 3$, means four. Random combinations of the analytic patterns give a population view of the same question. We drew 400 combinations per cap, with complex Gaussian coefficients on v_0 to v_7 whose standard deviation falls as $1/(j+1)$. The median τ for $kR \ge 3$ is 0.96 to 0.98 on caps with $\theta_0 \le 90^\circ$, 0.83 at 120° and 0.76 on the nearly closed cap. The same measure applied to the iterates of the six design-point runs gives 0.92 along the sinc run from the uniform thickness and 0.99 along the exact run. Pooled over all six runs, the

path designs within 3 dB of X0 (1020 iterates) reach only 0.02, those within 3 dB of SX (996) −0.02, and those within 1 dB of X0 (845) turn negative, −0.42. The iterates are successive designs rather than independent samples, so these values carry no significance level.

Appendix F. Face power at 100 Hz

The plateau near 0.5 suggests a picture: both faces radiate, and the sinc model counts one. Testing it needs a definition of face power, and two are in use; Fig. F.1(a) draws the first. The first is the *surface acoustic work* that each face does on the fluid. With $\mathbf{n}$ pointing to the outer face and v_n positive along $\mathbf{n}$, the outer face moves with velocity v_n against p^+ and the inner face with $-v_n$ against p^-:

$$P^+ = \tfrac{1}{2}\,\mathrm{Re}\int_\Gamma p^+\,\overline{v_n}\,dS, \qquad P^- = -\tfrac{1}{2}\,\mathrm{Re}\int_\Gamma p^-\,\overline{v_n}\,dS, \qquad P^+ + P^- = P_{2s}, \tag{F.1}$$

where the sum follows from Eq. (4). A face with negative work absorbs power from the fluid. The second is a *far-field angular partition*: the power radiated into the far-field directions with a positive axial component (the convex side) or a negative one. It cannot be negative, and it does not say which face did the work.

Measured on the coarse boundary-element mesh, the outer face does 69% of the surface work at 100 Hz for the uniform design, 62% for design G and 43–50% for designs B to E. The sinc power computed on the same mesh is 0.92 to 1.14 times the outer-face work, and the two follow each other from design to design with a correlation of 0.98; both are normalized by the same boundary-element power, which inflates that correlation. The far-field partition of design G, in contrast, is nearly even, 0.506 to 0.494, and that of the sinc pattern is 0.521 to 0.479; for the uniform design the far-field convex half is 0.502. For this one coarse mesh the sinc power is thus close to the outer-face work and differs from the far-field half split, which is consistent with a one-face picture without establishing it generally. Away from 100 Hz the match breaks down. At $kR = 1$ the inner face of design G absorbs power ($P^-/P_{2s} = -4.7$, $P^+/P_{2s} = 5.7$) while the sinc model over-predicts by a factor of 15; at $kR = 4$ the work divides about 40 to 60% between the faces and the mesh sinc power falls to 0.25 of the boundary-element far-field power; at $kR = 10$ the mesh sinc power is 1.26 times the outer-face work.

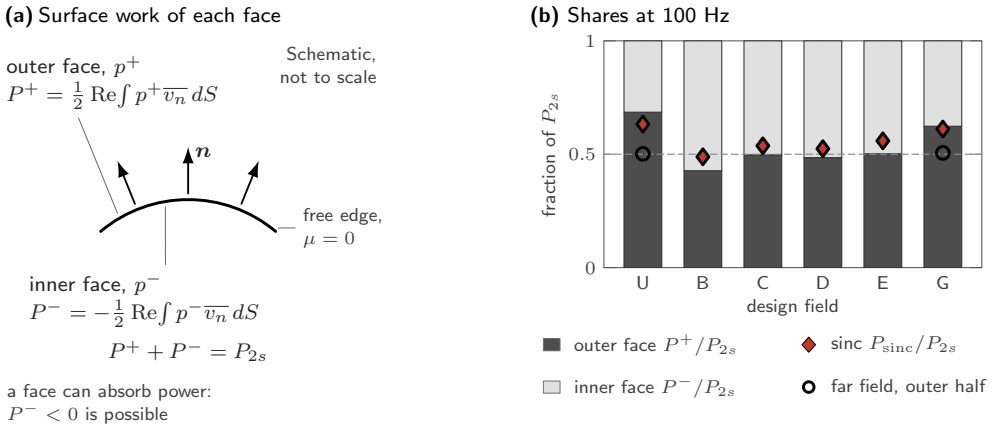


Figure F.1: Face power of an open cap at 100 Hz. (a) Surface acoustic work of each face, with **n** pointing to the outer face and the sign convention of Eq. (F.1); a face with negative work absorbs power; schematic, not to scale. (b) Measured shares of P_{2s} on the coarse boundary-element mesh for the six design fields: dark bars, outer-face work P^+; light bars, inner-face work P^-; diamonds, sinc power on the boundary-element mesh over P_{2s}; open circles, far-field convex half (U and G only), the second definition of the text. The dashed line marks one half.

Appendix G. Regime study: scaling, gaps, warning signs and descent directions

The regime study scales its operators as described in Section 2.4, except in the four design-point runs taken from Table 2, which keep the unscaled operators; the scaling keeps the divisor p_0 of Eq. (5) positive. Warm starts were rescaled to J_{ref} likewise after one unscaled warm run, started at 3.8×10^{-11} W, raised the power by 57 dB; that run was set aside. The evaluation set of a case held 647 to 767 designs, and Fig. G.1 shows two of the low-kR designs.

In the order of the rows of Table 3, the two exact-driven runs of each case ended 8.5, 0.4, 3.1, 5.8, 5.3, 5.9 and 8.8 dB apart in exact power at the 600-iteration limit (1.8 to 2.8 dB after 3000 iterations in the three cases continued, Section 4.4), and the ERP designs from the uniform start lie 53.5, 47.7, 53.1, 61.8, 39.8, 37.9 and 12.7 dB above the best exact design (gaps s). Figure G.2 uses the same layout as Fig. 7. In panel (b), every ERP run predicts about −15 dB regardless of its true level, while the exact runs lie on the equal-prediction line by construction.

In the asymptotic small-kR limit, the two faces of the open shell move

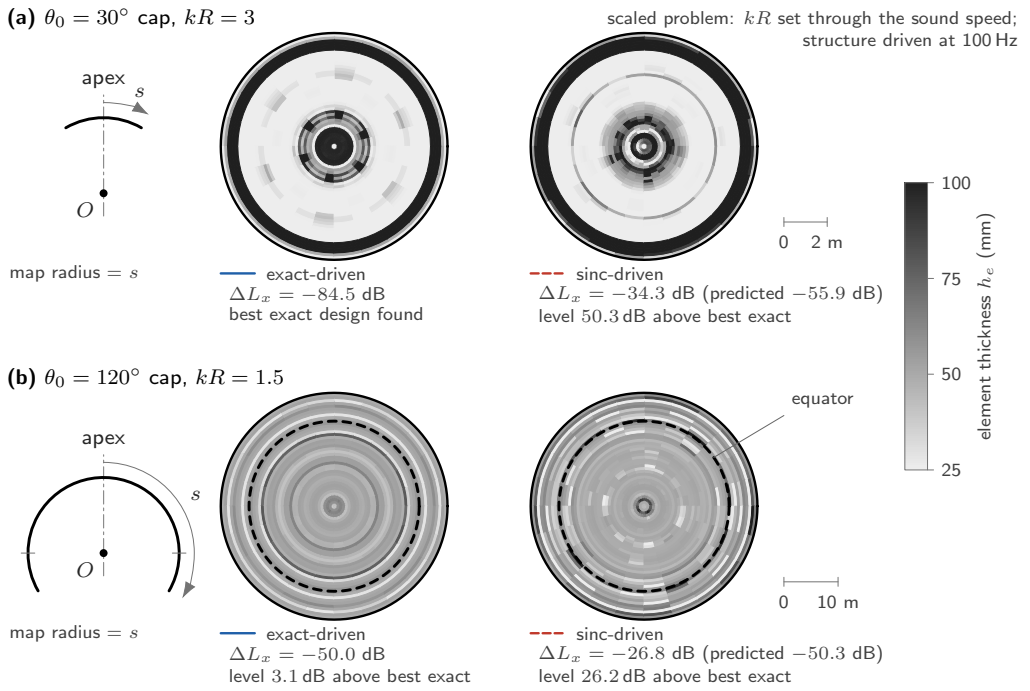


Figure G.1: Optimized thickness designs in two scaled low-kR cases, both from the uniform 50 mm design. (a) 30° cap at $kR = 3$; (b) 120° cap at $kR = 1.5$. Left, exact-driven; right, sinc-driven, each with its true level ΔL_x relative to the uniform design and its level above the best exact design. Each element is drawn flat on a map of the cap centered on the apex, whose radius is the meridian arc length; one gray scale from 25 to 100 mm. Scaled problem: kR set through the sound speed, structure driven at 100 Hz.

together and radiate with dipole-like behavior. At finite kR, the measured sinc-to-exact ratios depend on the velocity pattern: they cross unity between $kR = 1$ and 2.5 and reach their minima at $kR = 3$ or 4. The quietest designs rely on cancellation between parts of the surface, which the piston model evaluates by a different rule. No gap threshold was fixed in advance. Below some level, a difference between objectives cannot be told apart from the optimizer itself: the exact objective, started from two designs, ended up to 8.8 dB apart, and a replicate of run X0 whose objective was scaled differently by 4×10^{-8} followed the original to 10^{-9} for 10 iterations and then drifted by up to 1% in power. A line at about 10 dB therefore separates model from optimizer; the sinc gaps leave an empty band between 9.5 and 26.2 dB, so any line inside that band sorts the cases alike. Rank agreement on random

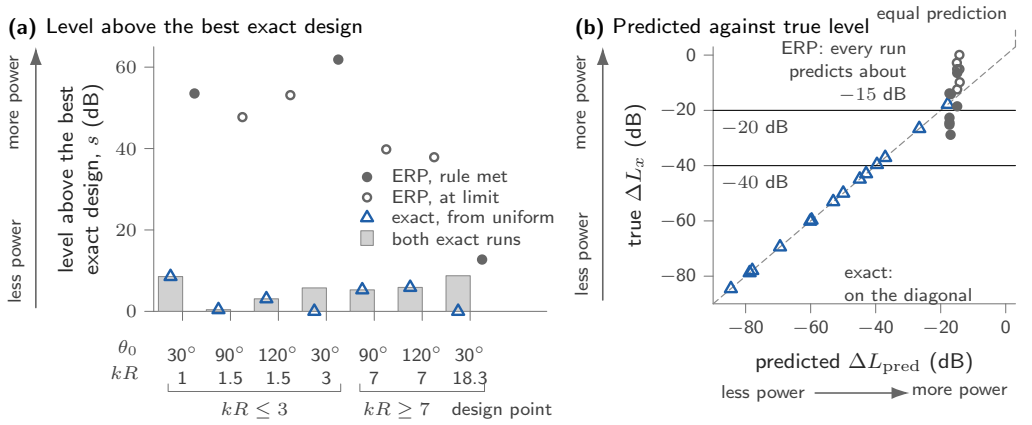


Figure G.2: ERP and exact-driven runs in the seven regime-study cases ordered by kR. (a) Lower-bound gap s above the best exact design for ERP runs from the uniform start (circles; filled: stopping rule met, open: 600-iteration limit); triangles show exact runs from the uniform start, and gray bars their spread. (b) Predicted versus true level relative to uniform for 14 ERP runs (circles) and 14 exact runs (triangles; equal by construction). Horizontal lines mark the example targets −20 and −40 dB. Conditions as in Fig. 7.

designs does not warn of failure: Kendall's τ between sinc and exact power was 0.70 to 0.89 in the failing cases and 0.92 to 0.98 in the others. Nor does the gradient at the start: the cosine between the projected sinc and exact gradients at the uniform design was 0.70 to 0.98 in the failing cases. Only the ratio ρ of the uniform design coincided with failure, 5.24, 0.36, 0.20 and 1.67 in the four failing cases against 0.58 to 0.64 in the other three; with seven cases it serves as a warning sign rather than a validated predictor. From the uniform design, a short sinc step lowered the exact power in every case by 65 to 111% of what the exact step achieved, whereas near the exact end designs the cosine between the two gradients fell to between −0.40 and 0.66: the directions agree at the start and part near the optima. The exact-driven designs concentrate their thickness changes in a few rings (Fig. G.1).

Appendix H. Benchmark cards

Figure H.1 sketches the three cases; Table H.1 summarizes their acoustic references and the evidence for the sinc model.

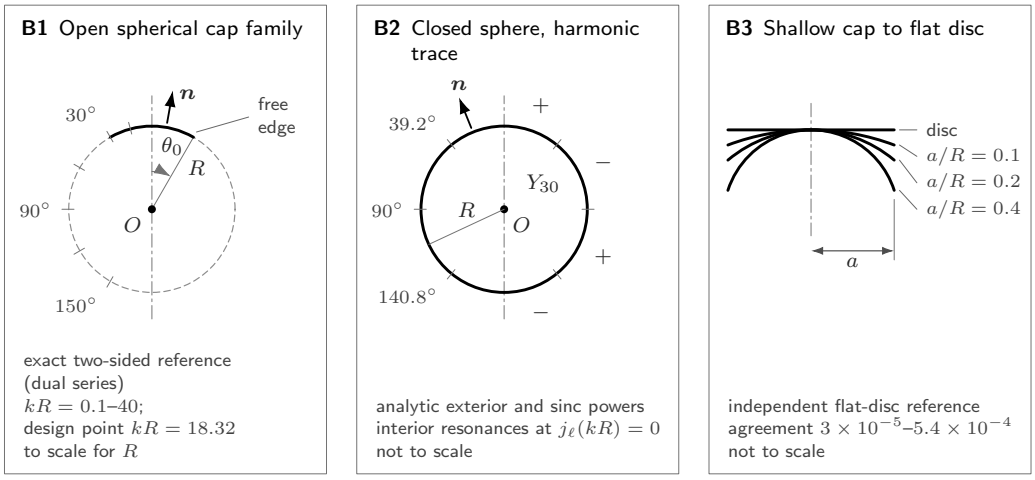


Figure H.1: Benchmark-case sketches. B1: an open spherical cap of radius R and half-angle θ_0; B2: a closed sphere with a spherical-harmonic trace; B3: a shallow cap of fixed rim radius a tending to a flat disc. The B2 analytic reference is not derived in this paper; the diagrams are schematic.

Card	Reference	Sinc model: measured, analytic or expected error
B1	exact two-sided dual series; BEM 0.1–0.6% on identical traces	measured (frozen fields): $\rho \gg 1$ at low kR, minima 0.15–0.25 at $kR = 3$–4, about 0.5 above $kR \approx 10$; 2× sinc within ±20% for 92% of points at $\theta_0 = 90°$, 18% at 150°; measured (optimization, $\theta_0 = 30$–120°): sinc-driven designs reach −26.8 to −41.9 dB at $kR = 1$–7 and −17.1 dB at $kR = 18.3$, 26–50 dB above the best exact design at $kR \leq 3$ (about 46 dB with the best exact level set at the −80 dB resolution limit; Section 4.2); with fluid loading in the scaled medium 18–36 dB above it, the in-vacuo exact optima rising by 2–15 dB (Section 6)
B2	analytic exterior and sinc powers per degree ℓ	analytic, not shown here: $\rho \to 2(\ell+1)^2/(2\ell+1)^2$ for subsonic traces; window mean 1 for supersonic traces; arbitrary error near $j_\ell(kR) = 0$
B3	independent flat-disc solution in the Hankel domain	measured: cap tends to disc as $(a/R)^2$; disc ratio is the flat-plate baffled-versus-unbaffled curve

Table H.1: Summary of reference methods and available evidence on sinc-model power for cases B1–B3. B1 measurements are finite-set results; see Sections 4.2 and 6 for the resolution-floor and fluid-loading qualifications.

Appendix I. Error populations and set-conditional budgets

Table I.1 gives the largest error over the three populations of evaluated designs discussed in Section 4.3. The random fields give their smallest maxima where the sinc model errs most, and the random-field maximum made the guaranteed level of the 30° cap at $kR = 1$ 34.0 dB too low, that is, too optimistic.

At the design point, the error spans less than 4 dB. At low kR, it grows as predicted levels fall, and the same-class envelope reaches the true boundary for the −20 dB target near a prediction of −44 dB (Fig. I.1).

Along the sinc paths at $kR \leq 3$ the error grows almost in proportion to the depth of the predicted level, with a correlation of 0.88 to 0.99. A constant budget, the largest error of the other cases of the class, fails in both

Table I.1: Largest error of the sinc model, $e_{\max}$ in dB, over three populations of evaluated designs, by class of case; $e = \Delta L_x - \Delta L_{\text{pred}}$ is positive when the design is louder than predicted. Ranges are over the four cases with $kR \leq 3$ and the two with $kR = 7$. Designs per case: 40 random smooth thickness fields; 242 exact-driven and 242 sinc-driven iterates (1803 and 1312 at the design point, whose random fields are those of the regime-study run of the same case). The last row gives the upper envelope of the sinc-driven iterates of each class, pooled over its cases.

Population	$kR \leq 3$	$kR = 7$	design point
random fields	0.03 to 2.63	0.08 to 0.22	0.51
exact-driven iterates	-18.2 to -5.3	0.27 to 1.11	0.31
sinc-driven iterates	19.0 to 35.2	1.27 to 1.45	0.61
envelope (sinc-driven)	$5.09 - 0.435\,\Delta L_{\text{pred}}$	$0.05 - 0.037\,\Delta L_{\text{pred}}$	

directions: it rejected four designs that reached the -20 dB target, and it made the guaranteed level of the 30° cap at $kR = 1$ up to 5.8 dB too low (too optimistic), because its predictions run lower than those of the cases it was measured on. At $kR \geq 7$ and at the design point the slopes are at most 0.05 and $e \leq 1.5$ dB.

The expected-value form adds little at the design point and hides the spread between starts. Only three sinc runs exist there, S0, SG and SX, and SX starts at the exact design, so their mean stands in for an expectation over starts without estimating one. Their predicted levels average -20.04 dB and their true levels -20.00 dB, equal to the -20 dB target. With the class envelope the guaranteed mean is -19.2 dB, and even the random-field maximum of 0.51 dB gives -19.5 dB: the mean misses that target, as do the two runs that did not start at the exact design. Averaging the powers of the three runs instead of their decibels gives -18.7 dB.

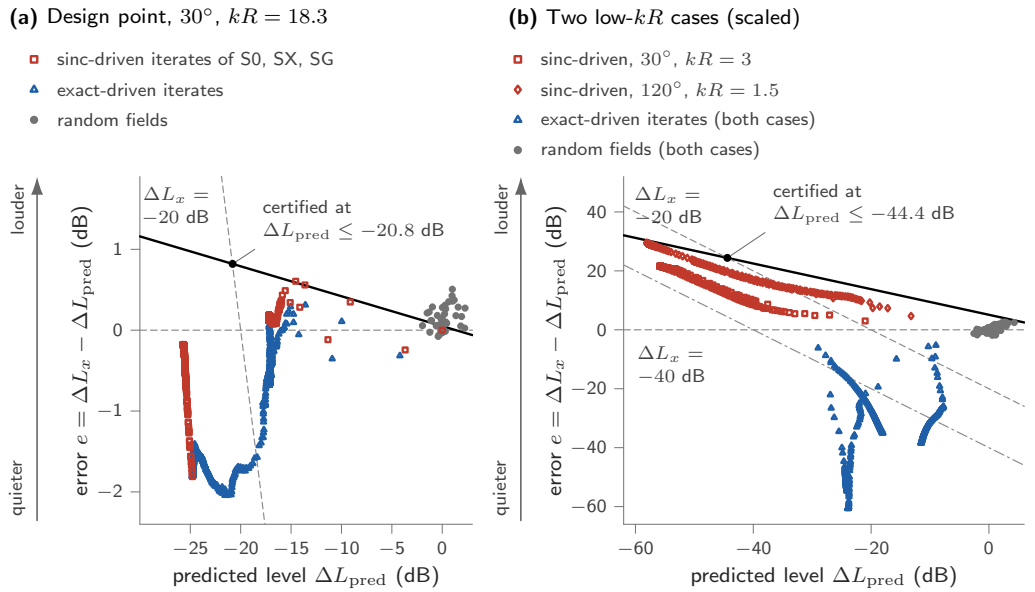


Figure I.1: Error of the sinc model, $e = \Delta L_x - \Delta L_{\text{pred}}$, against its predicted level ΔL_{pred} (relative to the uniform design, 0 at the right) for every evaluated design; positive e means that the design is louder than predicted. (a) Design point (30° cap, $kR = 18.3$): every iterate of the sinc-driven runs S0, SX and SG and of the exact-driven runs, and 40 random smooth thickness fields. (b) Two scaled cases at low kR, the 30° cap at $kR = 3$ and the 120° cap at $kR = 1.5$: every fifth iterate and the end designs of the sinc- and exact-driven runs, and 40 random fields each. Solid black line: upper envelope of the sinc-driven iterates of the class ($kR \geq 7$ in (a), $kR \leq 3$ in (b)). Dashed and dash-dot lines show the example targets -20 dB (both panels) and -40 dB (panel b); for any target T, the true-level boundary is $e = -T - \Delta L_{\text{pred}}$, and points below it meet the target. The dot marks where the envelope meets the $T = 20$ dB example line, the highest predicted level at which the rule certifies that example. Note the different scales of the two panels.

Appendix J. Longer runs and staged strategies: settings and results

The continued runs of Section 4.4 use the evaluator and the optimizer of the regime study, with one addition: the whole GCMMA state (design, asymptotes, the two previous designs, conservatism parameters, the divisor p_0 and the window of the stopping rule) is saved every 600 iterations, so that a run can go on in chunks. The new code gives the same objective and gradient as the regime-study code to the last bit, reproduces the first 30 iterations of a stored run bitwise, and a run split into two chunks equals an uninterrupted

one bitwise; the plain exact runs below also reproduce the 600 iterations of the regime study (-84.526 and -69.417 dB). The state at iteration 600 of the regime-study runs had not been stored, so each continuation restarts GCMMA from the returned design with fresh asymptotes and the objective rescaled to J_{ref}, a perturbation that released 5.7 dB on the 120° cap. Every level quoted here is the lowest over designs that satisfy the volume constraint to a relative 10^{-5}. The whole experiment took about 13 h of one MATLAB session with two threads, at 0.5 to 2.7 s per outer iteration.

The strategies were fixed before any of them was run, and nothing was tuned afterwards. Each starts from the uniform design with 1200 outer iterations, and each change of model is a warm restart under the rule of Appendix G. P is plain exact optimization. S1 runs ERP for 150 iterations, sinc for 350 and exact for 700; S2 runs sinc for 300 and exact for 900; S3 is S2 with a move limit of 5 mm per iteration, one fifteenth of the bound range. S4 is a continuation in kR: the sinc model at the wavenumber of the design point, $kR = 18.3$, for 300 iterations, then the exact model at $kR = 3$ for 900 (on the cap at $kR = 1$: 200 iterations of sinc at $kR = 18.3$, 300 of exact at $kR = 3$ and 700 of exact at $kR = 1$). S5 starts with the sinc model and switches to exact when the cosine between the projected sinc and exact gradients stays below 0.9 for three iterations, when the exact power rises above its value ten iterations earlier, or after 600 sinc iterations. Table J.1 gives the result.

At $kR = 3$ plain exact optimization reached -84.2 dB within 300 iterations, a level that no staged strategy reached within 1200 iterations or within the same measured optimization-loop time (operator construction excluded); since this level lies below what the reference resolves, the ranking was checked with the factor $\ell_{\max} = 8000$ alone, under which the final design of P still lies 1.6 dB lower than the best staged one. At $kR = 1$ the sinc stages helped early: S2, S3, S5 and S4 reached -69.4 dB, the level of P at iteration 600, after 349, 355, 423 and 474 iterations (189 to 226 s of optimization-loop time against 309 s, excluding operator construction), and at iteration 600 they lay 5.3 to 12.3 dB lower than P. P then fell below -80 dB by iteration 900 (-90.4 dB, a depth the reference does not resolve) and stopped at iteration 1076, lower than every staged strategy at both budgets; its final design is also the lowest under the factor $\ell_{\max} = 8000$ alone, by 2.8 to 3.8 dB against S2 to S5. A plausible mechanism, not tested here, is that the sinc stage places the design in a basin chosen by its monopole-like rating, which the exact stage then polishes without reaching the deeper basins that plain exact optimization finds.

Table J.1: Staged strategies on the 30° cap at $kR = 3$ and 1, from the uniform design: lowest true level ΔL_x reached by a feasible design within 300, 600, 900 and 1200 iterations, measured optimization-loop time (operator construction excluded), and the final design evaluated with the factor $\ell_{\max} = 8000$ alone. Levels in dB relative to the uniform design; levels below about −80 dB are not resolved by the reference (Section 4.4). Stages: model (E, ERP; S, sinc; X, exact) and number of iterations. One start per strategy; the plain exact runs met the stopping rule at iterations 1100 and 1076.

	stages	300	600	900	1200	time (s)	$\ell_{\max} = 8000$
30° cap, $kR = 3$							
P	X 1100	−84.22	−84.53	−84.77	−84.93	1717	−80.76
S1	E 150, S 350, X 700	−48.22	−68.70	−70.81	−71.14	1222	−71.21
S2	S 300, X 900	−33.48	−76.34	−78.60	−79.96	1624	−78.61
S3	as S2, move limit	−33.48	−77.17	−79.62	−80.85	2101	−79.18
S4	kR continuation	−23.90	−64.72	−69.89	−72.65	1002	−72.46
S5	switch after S 4	−65.08	−65.08	−65.08	−65.08	1619	—
30° cap, $kR = 1$							
P	X 1076	−55.90	−69.42	−90.40	−94.31	752	−82.51
S1	E 150, S 350, X 700	−24.33	−64.07	−72.10	−72.97	749	−72.90
S2	S 300, X 900	−36.52	−74.69	−77.15	−78.96	814	−78.74
S3	as S2, move limit	−36.52	−74.94	−77.50	−79.35	1129	−79.01
S4	kR continuation	−49.49	−79.77	−80.34	−80.86	837	−79.71
S5	switch after S 2	−61.71	−81.72	−81.79	−81.79	926	−79.65

The switching rule left the sinc model after 4 and 2 iterations, so S5 behaves like plain exact optimization restarted early. In both cases S5 then moved to designs that violate the volume constraint (by up to 0.34% at $kR = 3$, from iteration 143 on), where it reached −88.4 and −96.6 dB; thinned uniformly back to the allowed volume, these designs give −87.7 and −69.7 dB. Such levels are outside the rules of the comparison and are not counted; the feasible values are those of Table J.1.

Appendix K. Fluid-loading operator: checks and data

The operator is assembled per azimuthal order as in Algorithm 1: its reactive part from $\rho_0 c_0 \, \mathbf{A}_m^{-1}$ and its resistive part from the right-hand form of Eq. (C.1), because the Hermitian part of $\rho_0 c_0 \, \mathbf{A}_m^{-1}$ computed directly is dominated by round-off in the high orders; symmetrized on the free coefficients, the coupled matrix of the re-optimization stays complex symmetric, so one factorization serves state and adjoint. Two independent implementations of

Eq. (11) gave the same power for one design to the six digits printed, and the balance of input, dissipated and radiated power closes to 2.2×10^{-10} of the input power for every design. Four checks support the fluid-loading operator $\mathbf{Z}$ itself. Its resistive part reproduces the power operator $\mathbf{F}_x^{\mathrm{H}}\mathbf{F}_x$ to 2×10^{-9} to 1.2×10^{-6} in relative Frobenius norm over the seven cases, the difference being the clipping of negative eigenvalues; $\mathbf{Z}$ is symmetric to 1.3×10^{-9}, as reciprocity requires. Extrapolated to a vanishing hole on a nearly closed cap, its modal impedance matches that of a closed sphere with fluid on both faces to 2.6×10^{-4} for the breathing mode and 9.4×10^{-3} for degrees one to three ($kR = 1$ and 1.5). For a shallow cap, the added mass tends to the value $\frac{8}{3}\rho_0 a^3$ of a rigid disc moving broadside, with a ratio of 0.998. Table K.1 lists the best exact ΔL of Fig. 10 for all seven cases, together with the gap s of the sinc design (the vertical distance between the two lines of the figure, so that the sinc level is $\Delta L + s$) and the gap of the better of the two sinc designs in each case.

Table K.1: Fluid-loaded re-evaluation of regime-study designs optimized in vacuo. All row entries are evaluations of those in-vacuo designs; footnote *a* reports a separate fluid-loaded exact re-optimization for 30°, $kR = 3$. ΔL_x: true level of the exact-driven design relative to the uniform design; s: level of the uniform-start sinc design above the in-vacuo-optimized exact design, Eq. (10); best sinc: the same gap for the quieter sinc design. The footnote's gap is instead to the separately re-optimized exact design. Readings: in vacuo (vac), scaled medium (scaled), air density (air); levels in dB within each medium.

		best exact ΔL_x			sinc s			best sinc s		
θ_0	kR	vac	scaled	air	vac	scaled	air	vac	scaled	air
30°	1	−78.0	−75.8	−54.9	36.0	34.0	15.0	36.0	34.0	15.0
90°	1.5	−60.1	−57.3	−39.7	31.7	29.1	14.2	28.3	25.6	9.5
120°	1.5	−53.1	−44.4	−23.5	26.2	17.6	1.6	24.4	15.8	1.0
30°	3[a]	−84.5	−70.0	−54.9	50.3	35.9	21.8	50.3	35.9	21.8
90°	7	−44.9	−43.5	−39.5	5.7	4.8	2.3	5.7	4.8	2.3
120°	7	−42.9	−41.8	−38.1	3.6	2.7	1.1	2.1	1.5	0.8
30°	18.3	−26.6	−26.5	−26.5	9.5	9.5	9.5	0.7	0.7	0.7

[a] The in-vacuo depth of the 30°, $kR = 3$ case lies inside the noise of the reference (about −80 to −84.5 dB). Separate from the row entries above, the fluid-loaded exact re-optimization reaches $\Delta L = -84.5$ dB in the scaled medium and −83.9 dB in air after 150 iterations. The gaps from the uniform-start sinc design, evaluated in each medium, to these re-optimized designs are $s = 50.4$ and 50.7 dB; the row gaps instead use the exact design optimized in vacuo.

CRediT authorship contribution statement

G. H. da Silva: Conceptualization, Software, Writing – original draft. **A. Lenzi**: Conceptualization, Supervision, Resources, Writing – review and editing.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude (Anthropic), Gemini (Google) and GPT (OpenAI) for programming, citation checking, drafting and manuscript review. The authors developed the research ideas and designed the numerical experiments. They reviewed and edited the manuscript as needed and take full responsibility for its published content.

Data availability

The code, scripts, stored text results and figure sources supporting the results are available at `https://github.com/dshguilherme/piston`. The binary results listed there (designs, operator factors and states, 2.9 GB) will be deposited in a public data repository, and its DOI will be added to this statement.

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